

(12) **United States Patent**
Zhang

(10) **Patent No.:** **US 12,411,311 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **CAMERA OPTICAL LENS**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 355 days.

(21) Appl. No.: **18/324,122**

(22) Filed: **May 25, 2023**

(65) **Prior Publication Data**
US 2024/0248284 A1 Jul. 25, 2024

(30) **Foreign Application Priority Data**
Jan. 19, 2023 (CN) 202310066885.4

(51) **Int. Cl.**
G02B 9/64 (2006.01)
G02B 13/00 (2006.01)

(52) **U.S. Cl.**
CPC **G02B 9/64** (2013.01); **G02B 13/0045**
(2013.01)

(58) **Field of Classification Search**
CPC .. G02B 9/64; G02B 13/0045; G02B 13/0015;
G02B 13/006; G02B 13/06; G02B 13/18
USPC 359/755
See application file for complete search history.

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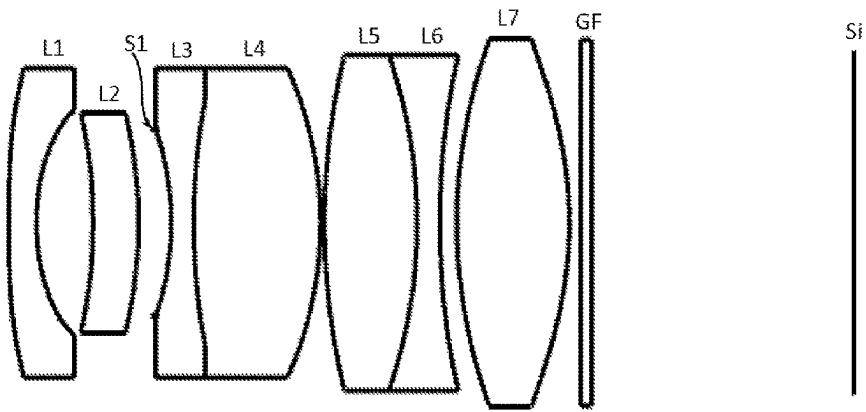
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(57) **ABSTRACT**

A camera optical lens is provided, including from an object side to an image side: a first lens; a second lens; a third lens; a fourth lens; a fifth lens; a sixth lens; and a seventh lens, the camera optical lens satisfies following conditions: $1.30 \leq f_7/f_4 \leq 4.00$; $1.70 \leq n_1$; $2.00 \leq f_{34}/f_7 \leq 7.00$; and $120.00 \leq (FOV \times f)/IH$, where f denotes a focal length of the camera optical lens; f_7 denotes a focal length of the seventh lens; f_{34} denotes a combined focal length of the third lens and the fourth lens; n_1 denotes a refractive index of the first lens; FOV denotes a field of view of the camera optical lens; and IH denotes an image height of the camera optical lens. The camera optical lens has good optical performance such as large aperture, ultra-thinness and wide angle while ensuring miniaturization, small chromatic aberration, small lateral color, and high resolution up to 5M.

16 Claims, 15 Drawing Sheets



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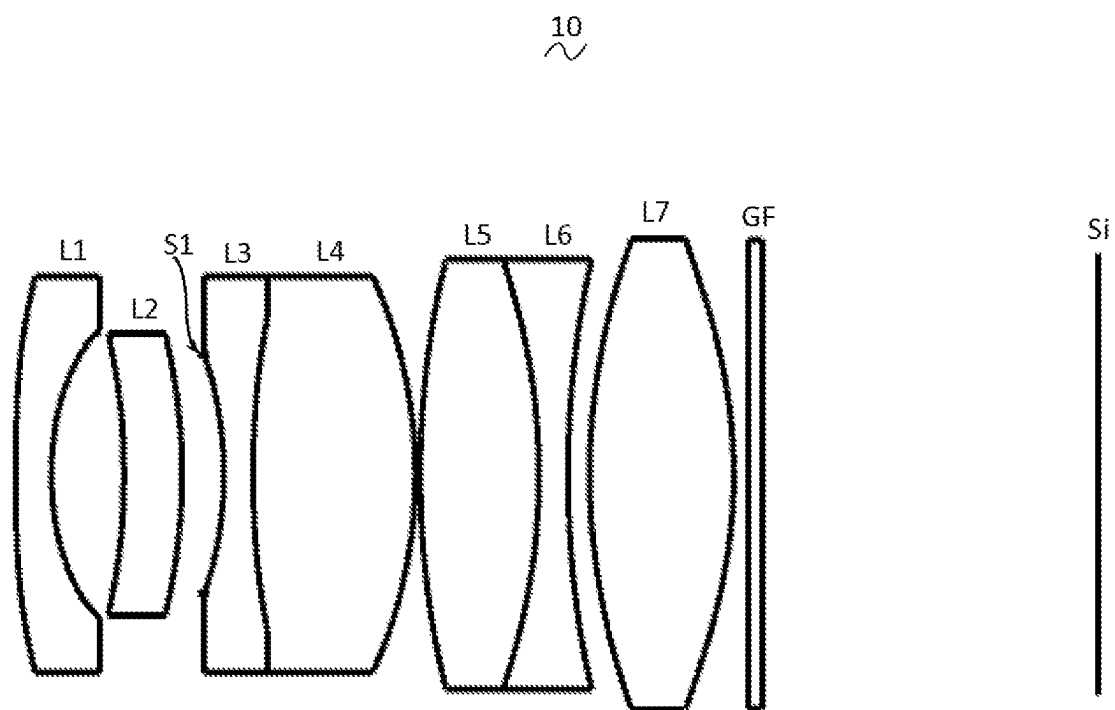


FIG. 1

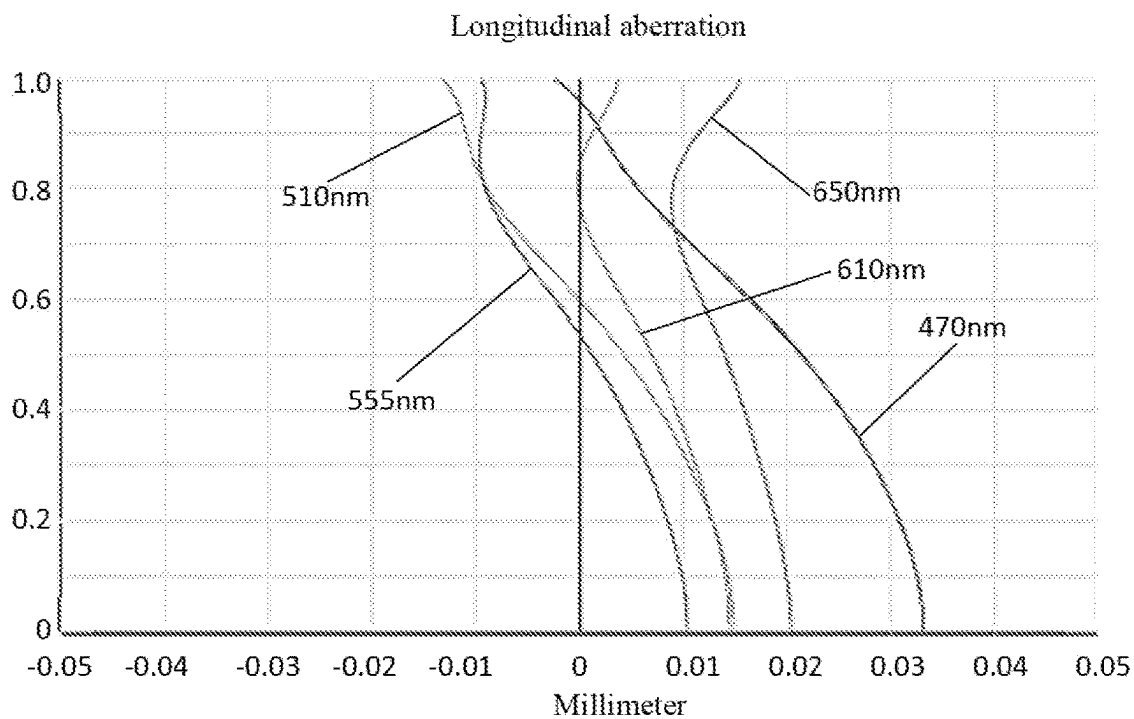


FIG. 2

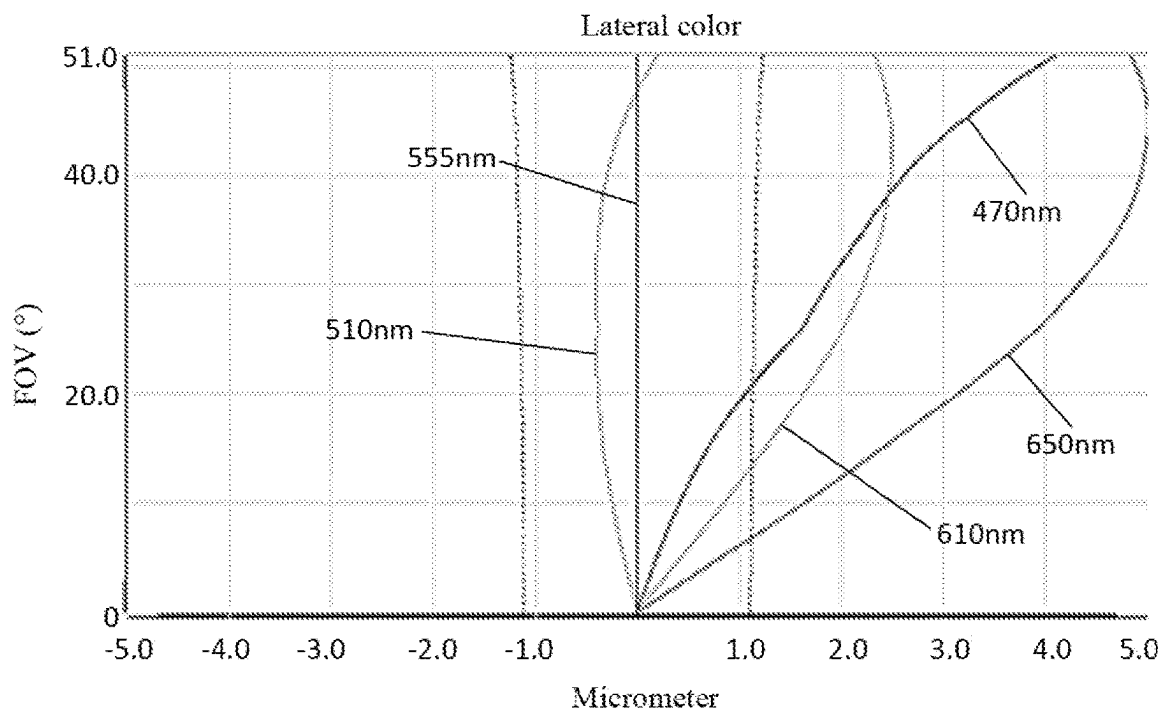


FIG. 3

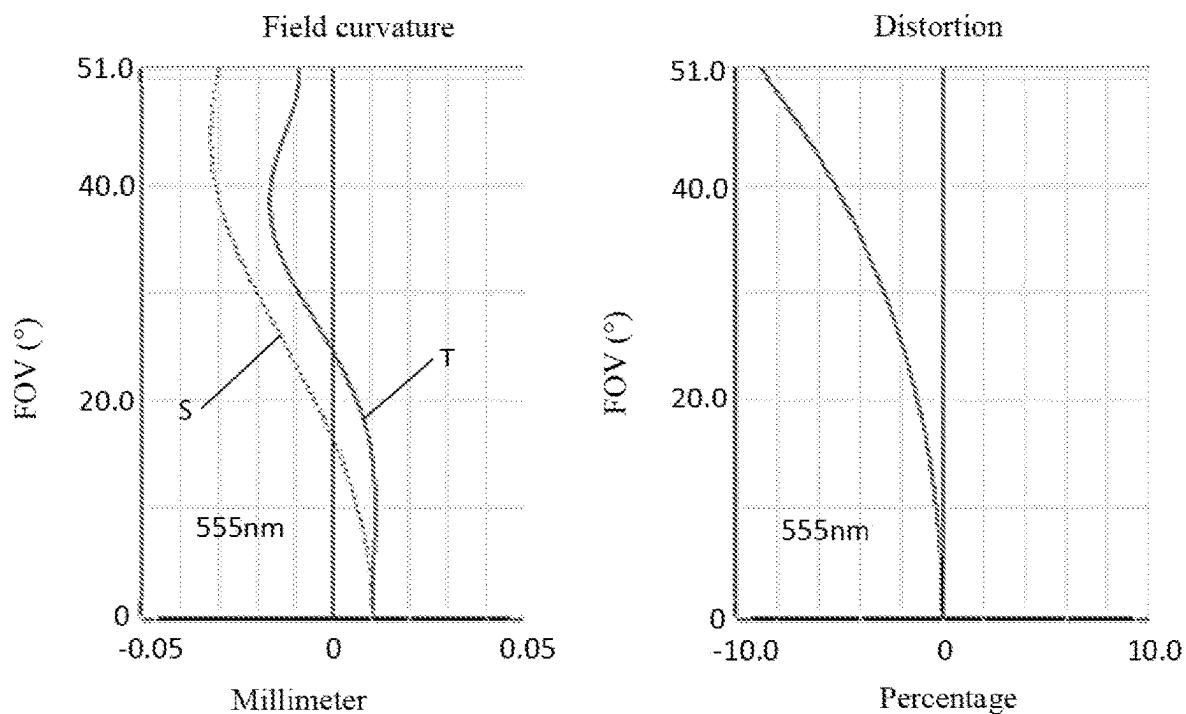


FIG. 4

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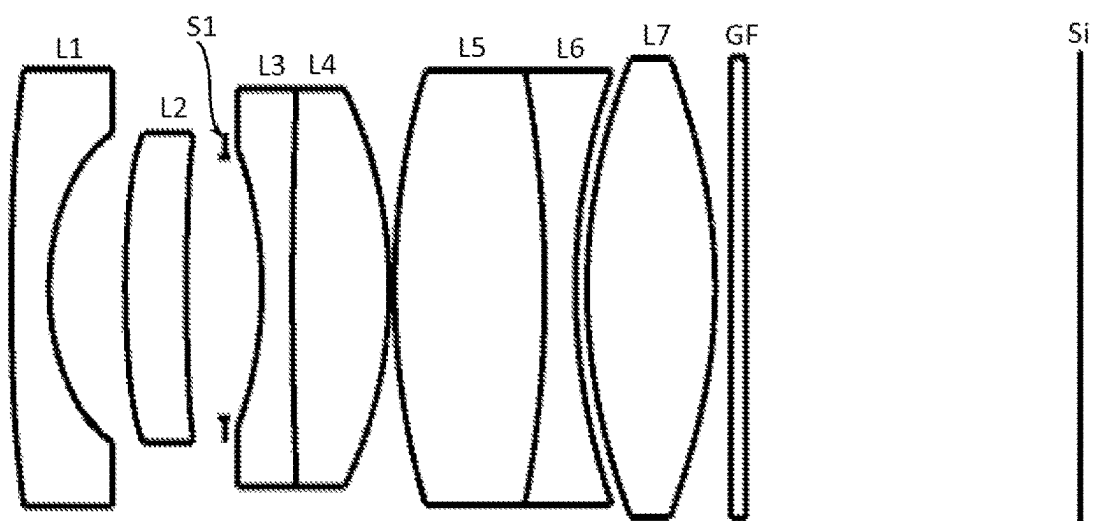


FIG. 5

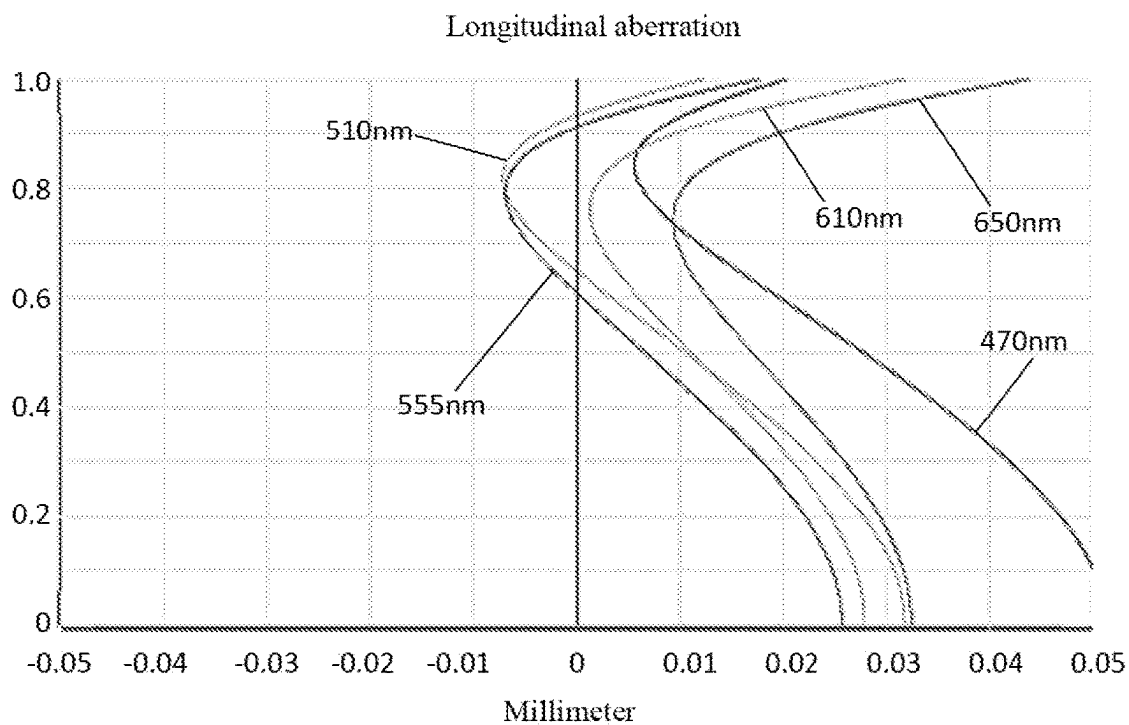


FIG. 6

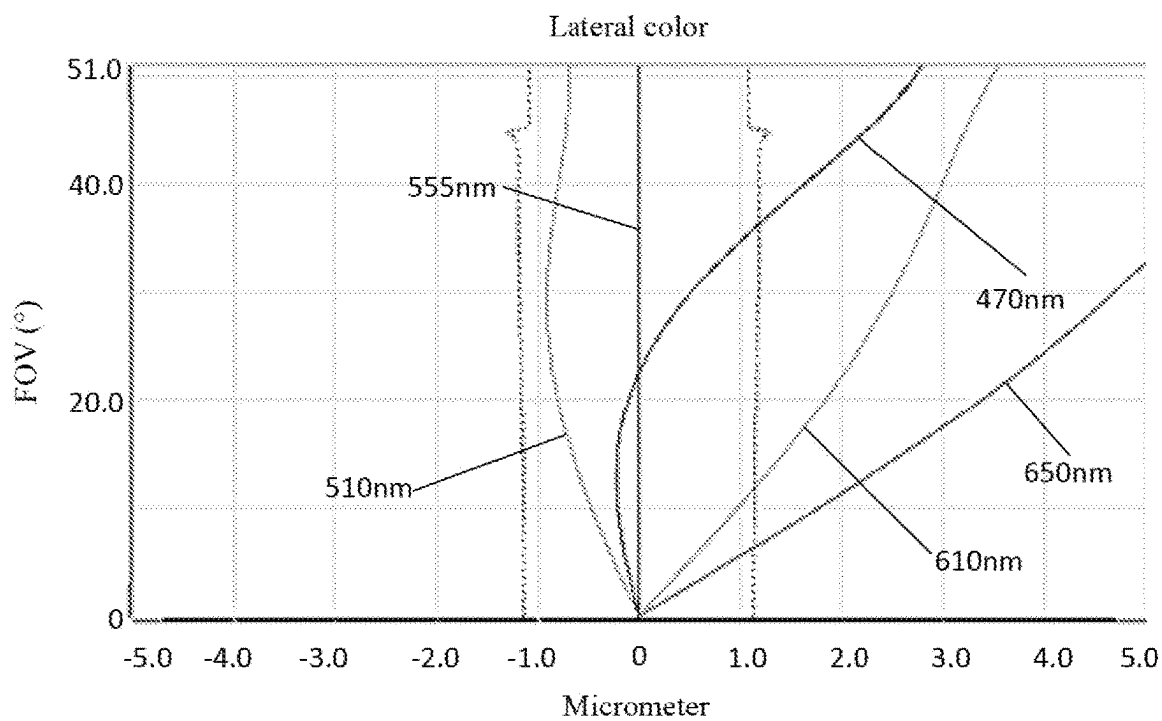


FIG. 7

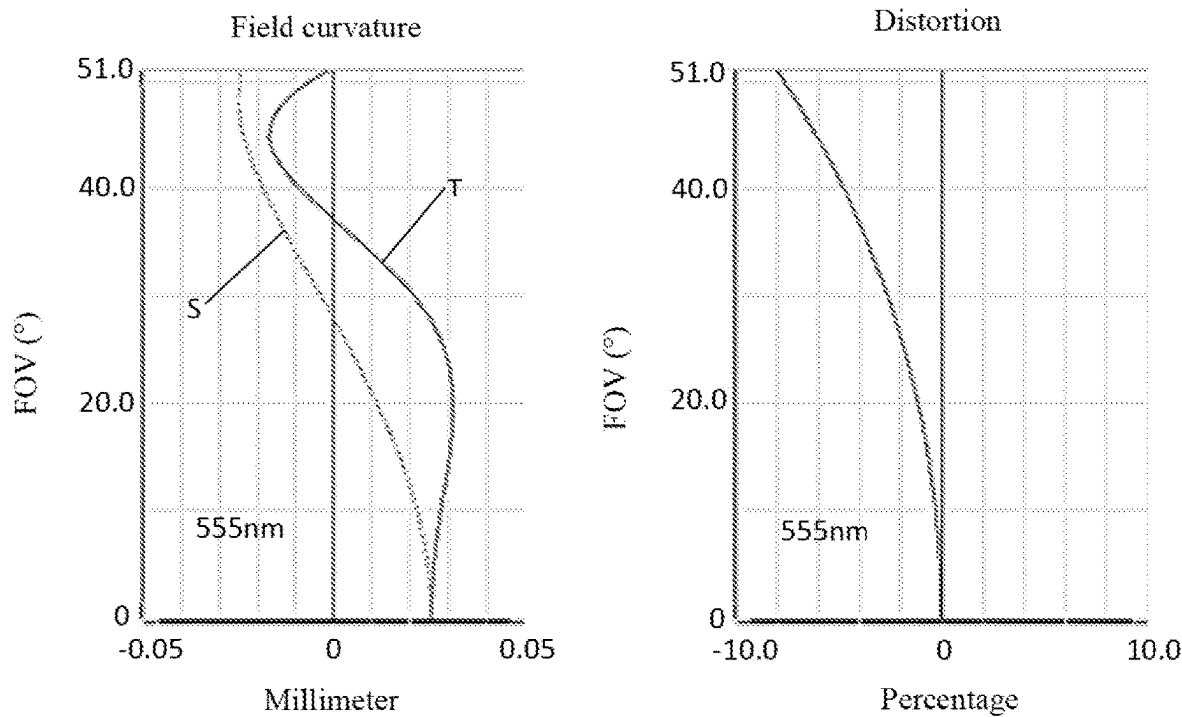


FIG. 8

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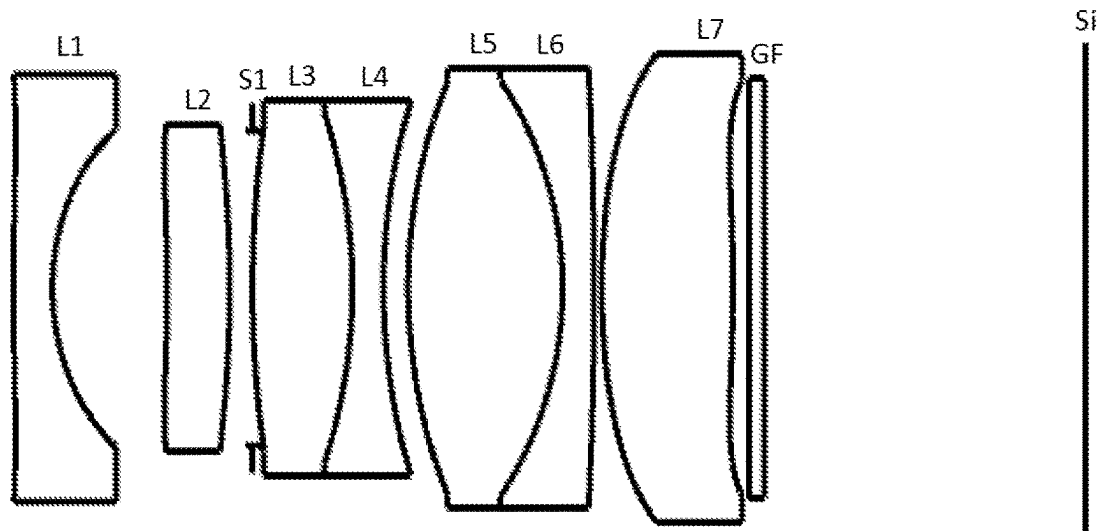


FIG. 9

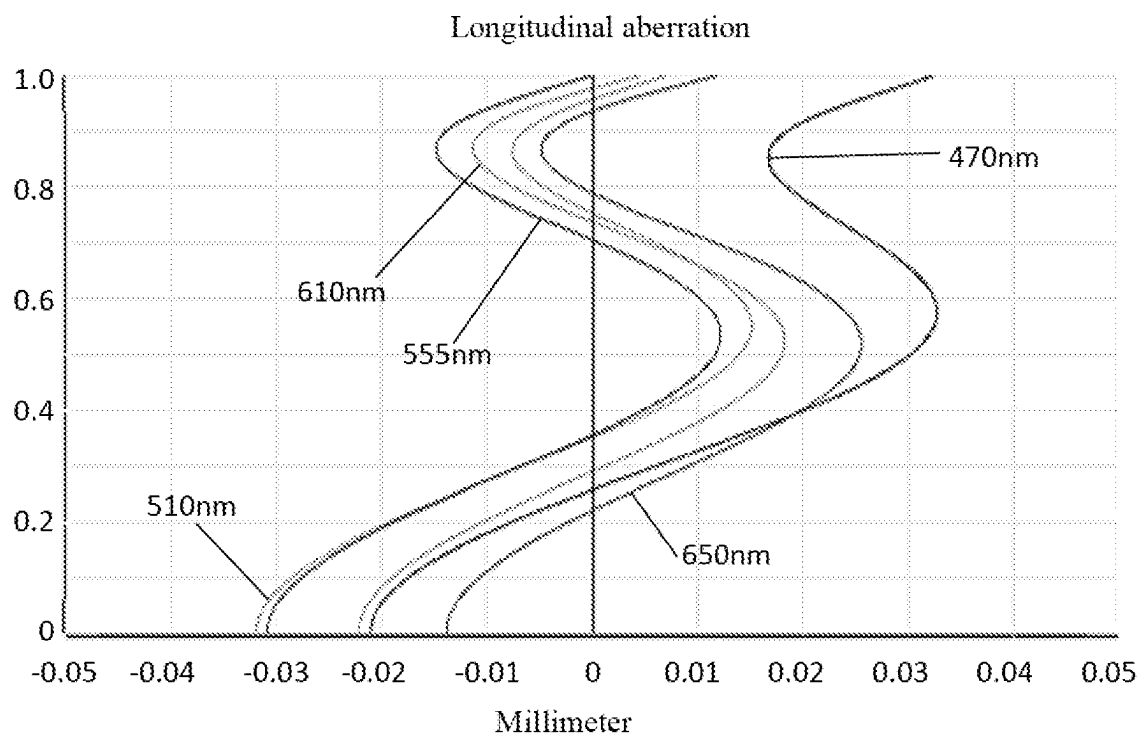


FIG. 10

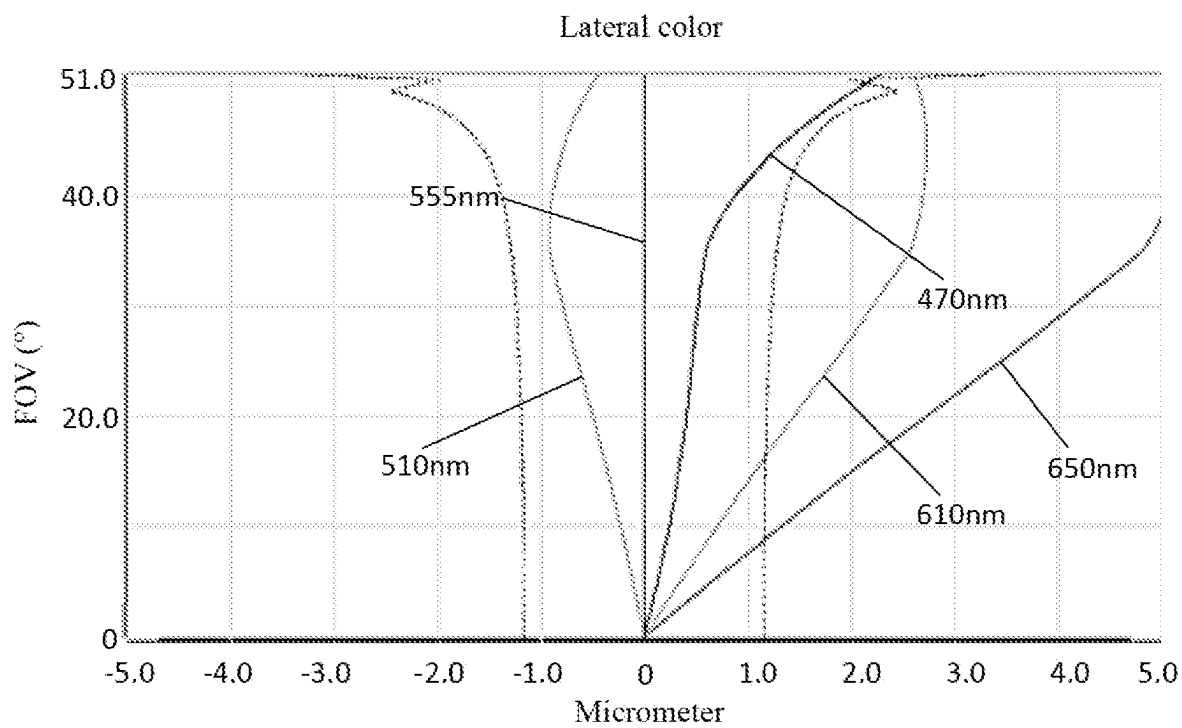


FIG. 11

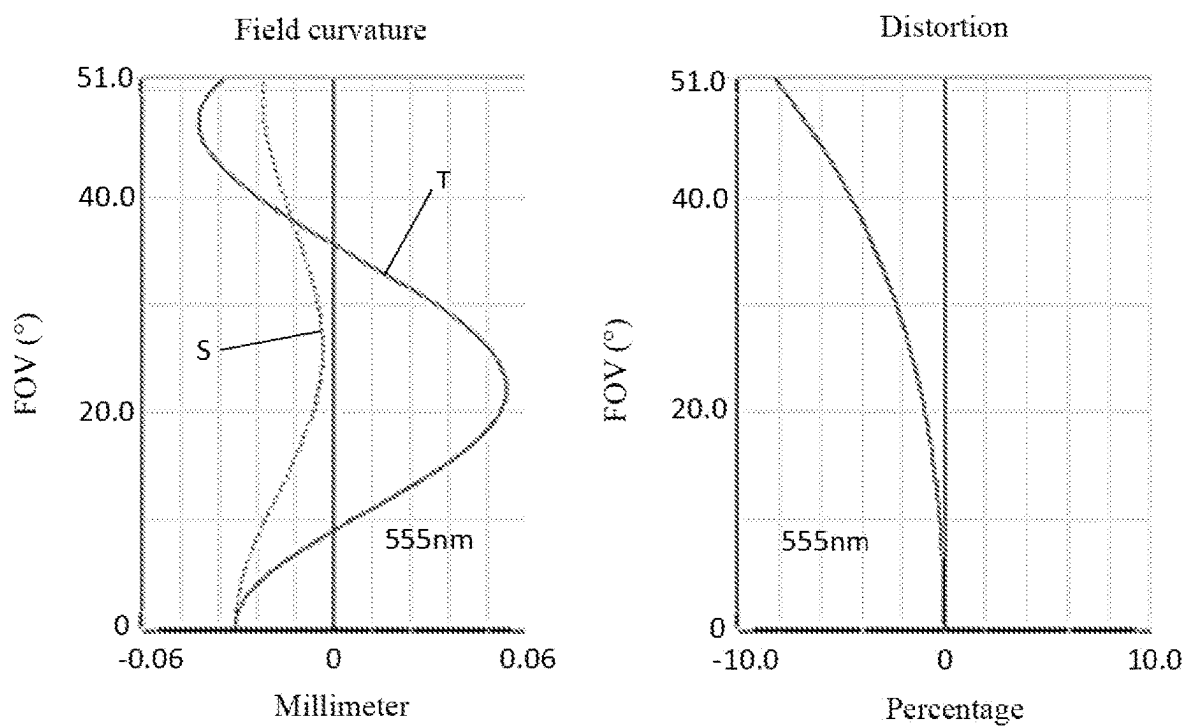


FIG. 12

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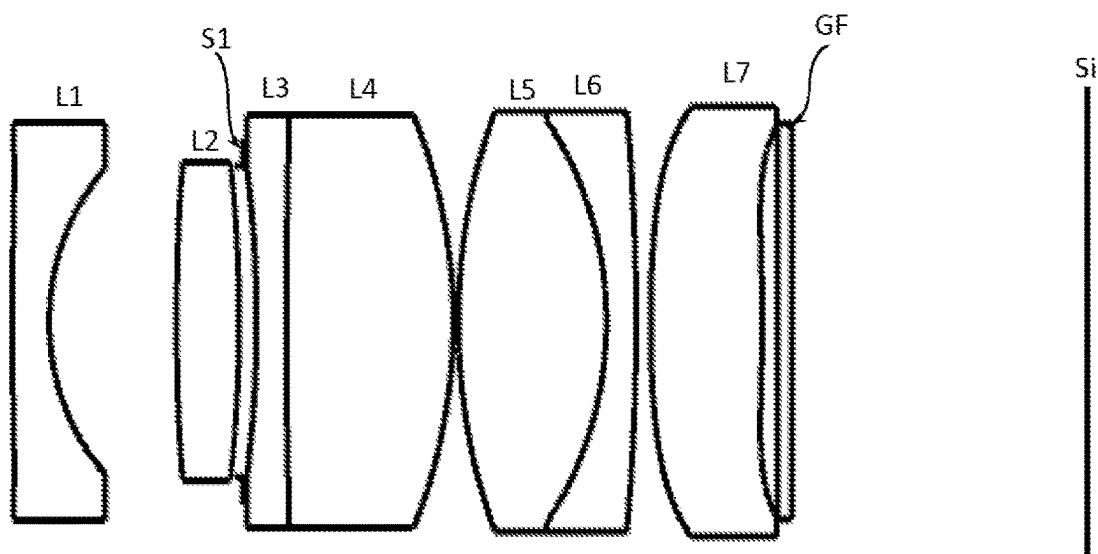


FIG. 13

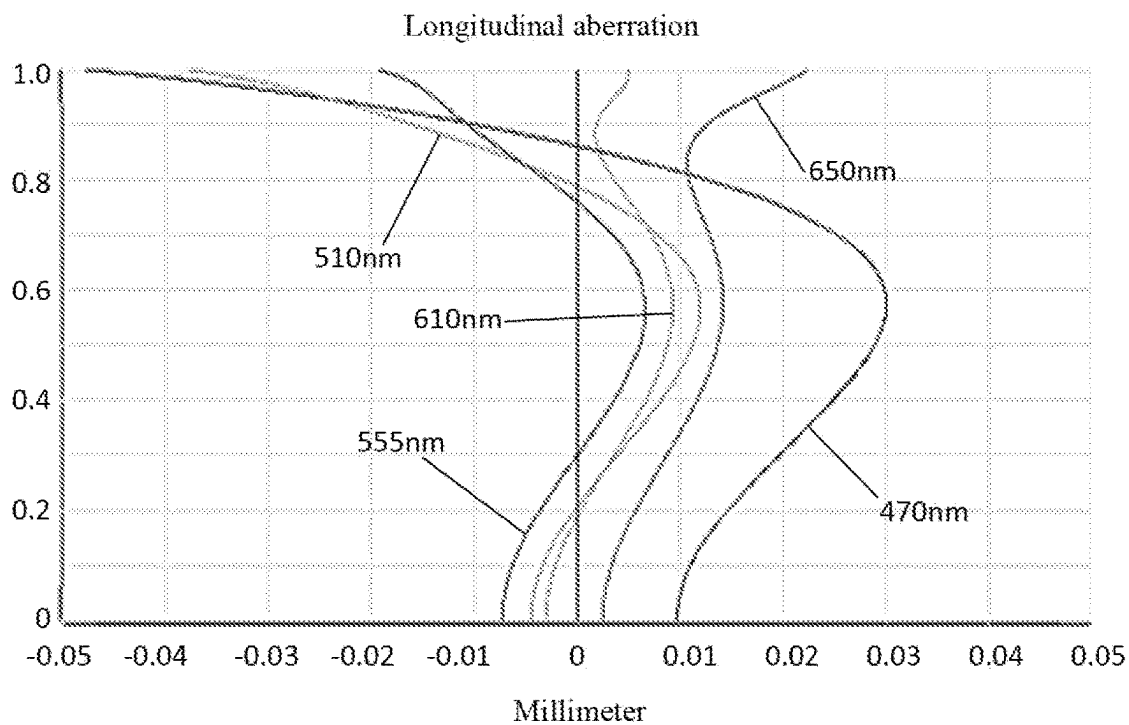


FIG. 14

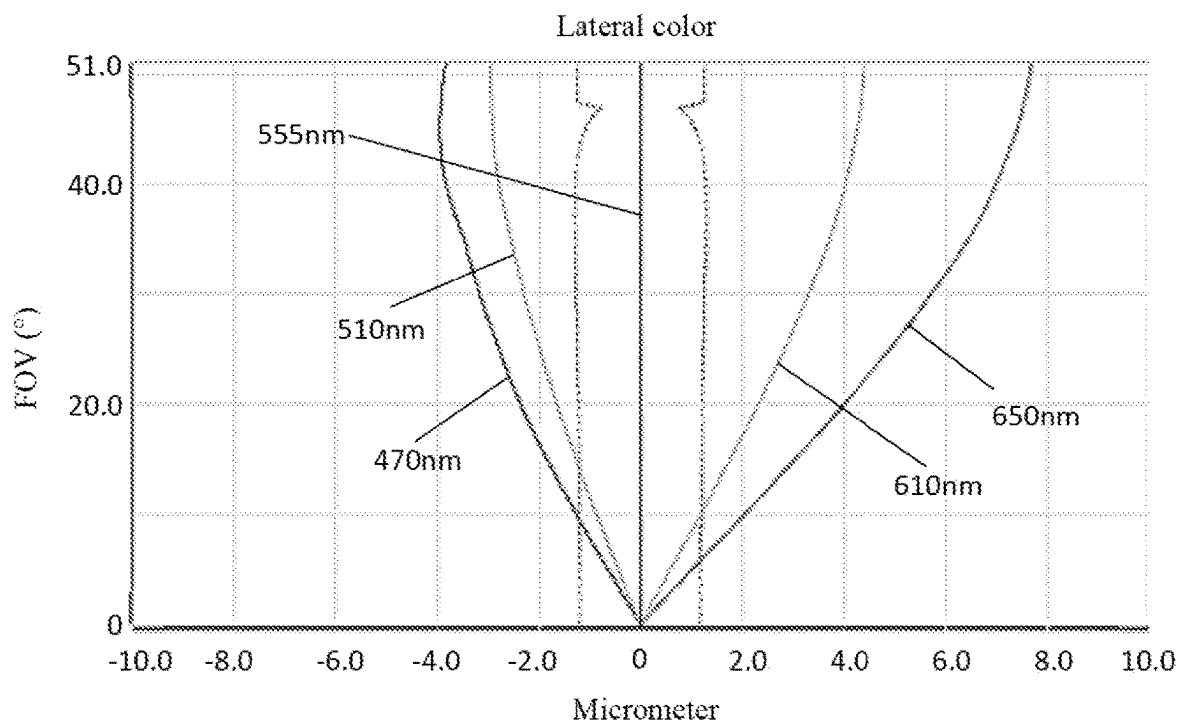


FIG. 15

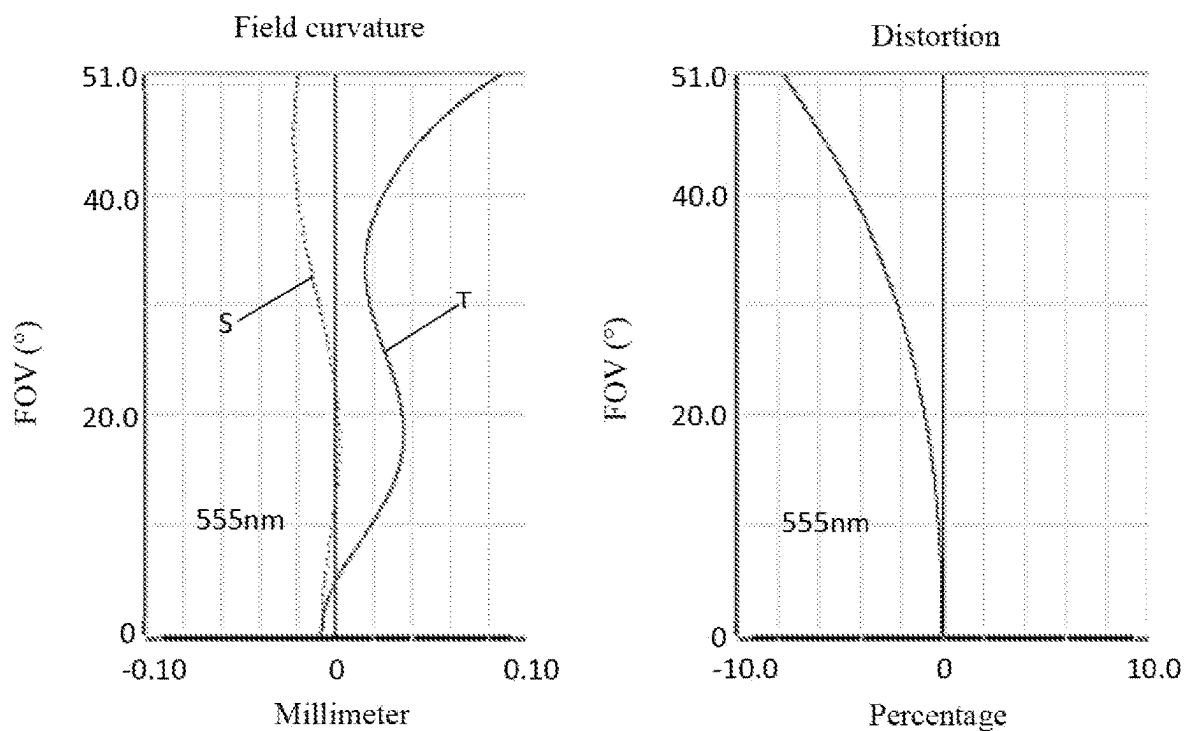


FIG. 16

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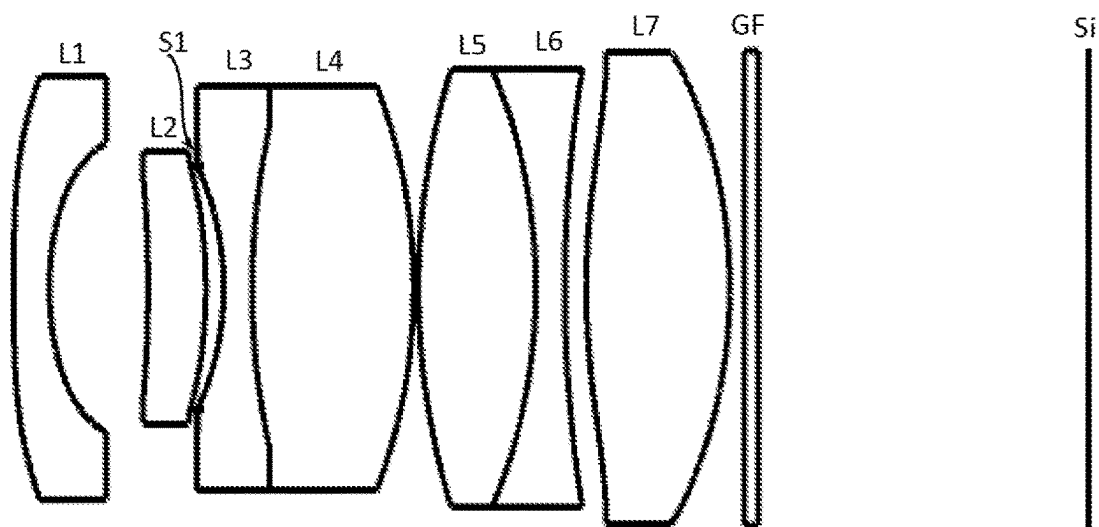


FIG. 17

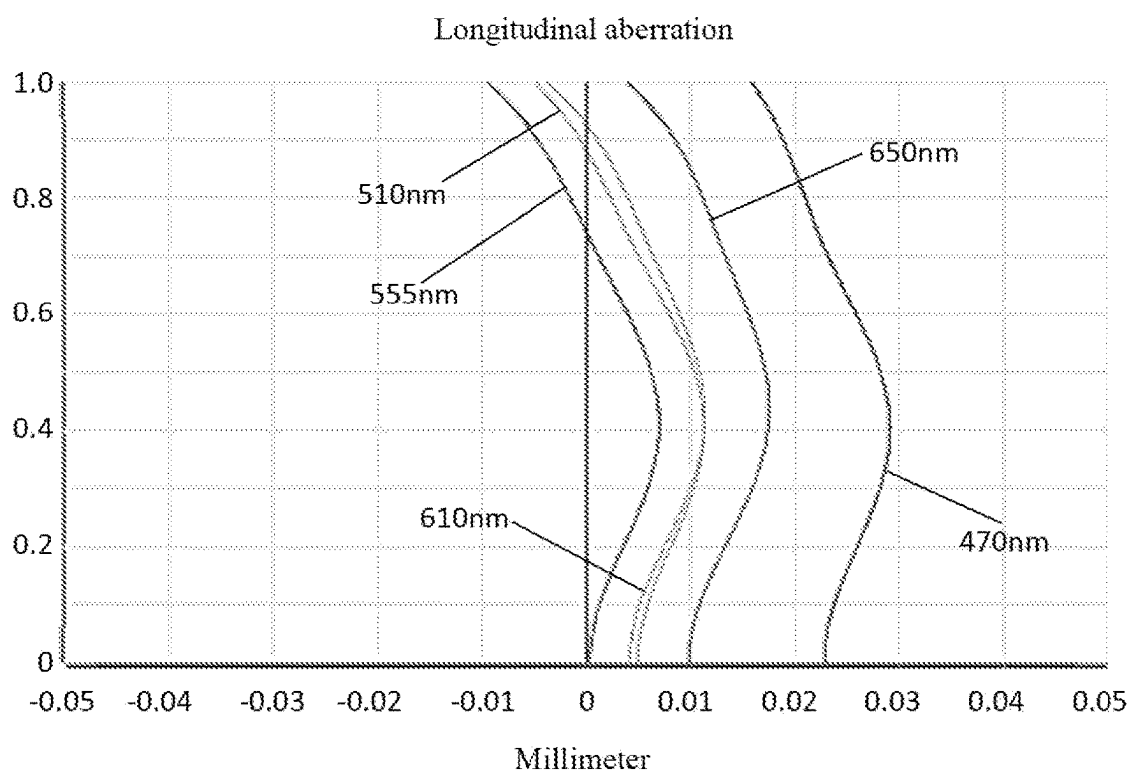


FIG. 18

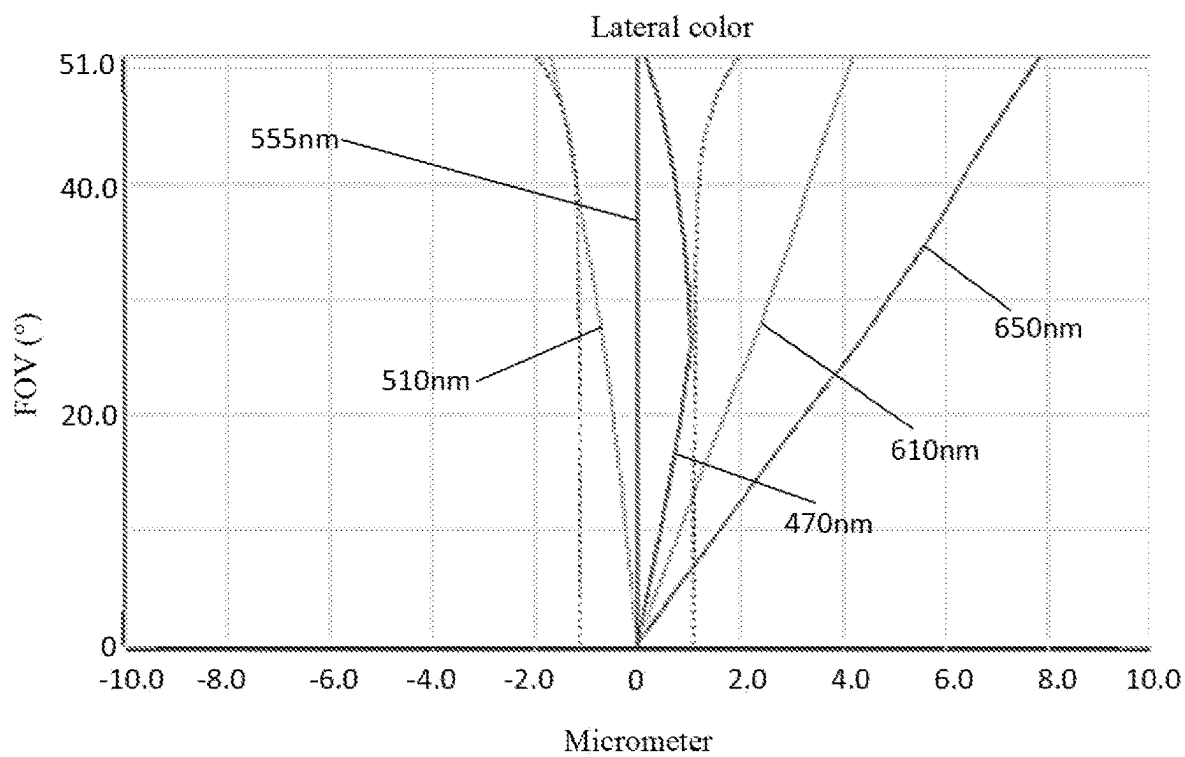


FIG. 19

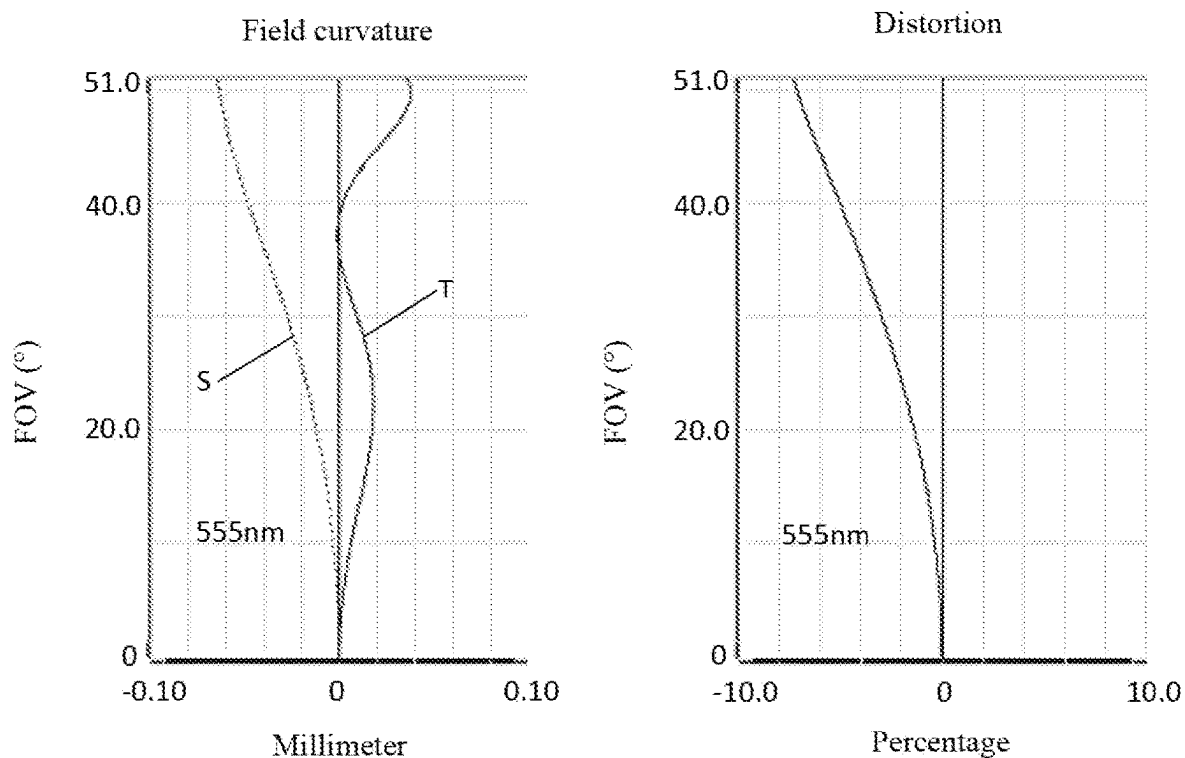


FIG. 20

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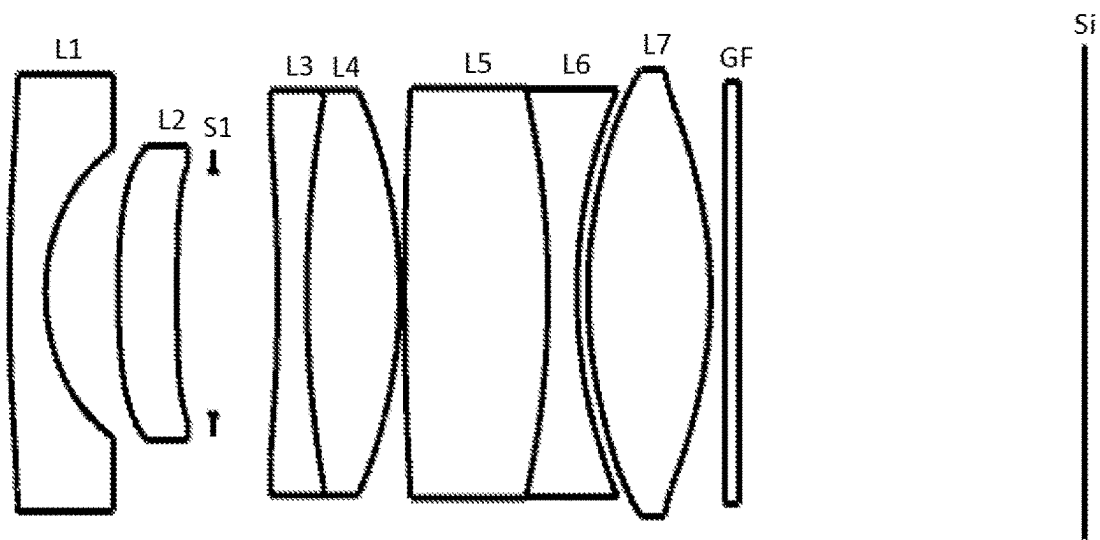


FIG. 21

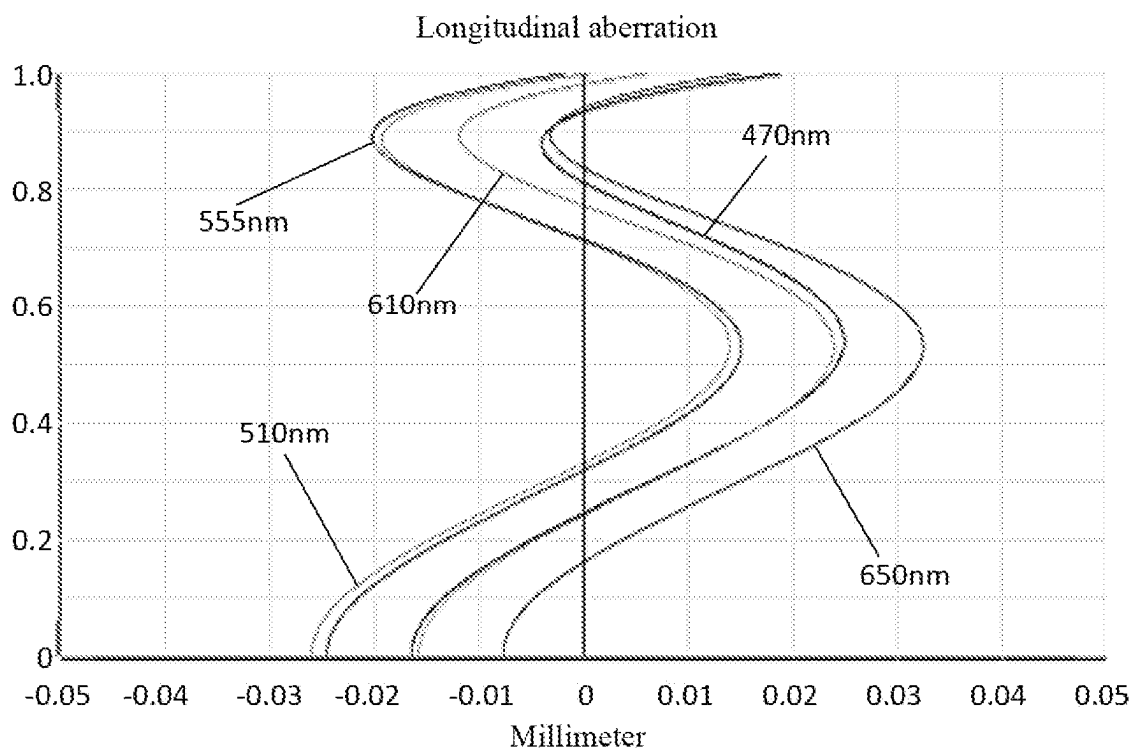


FIG. 22

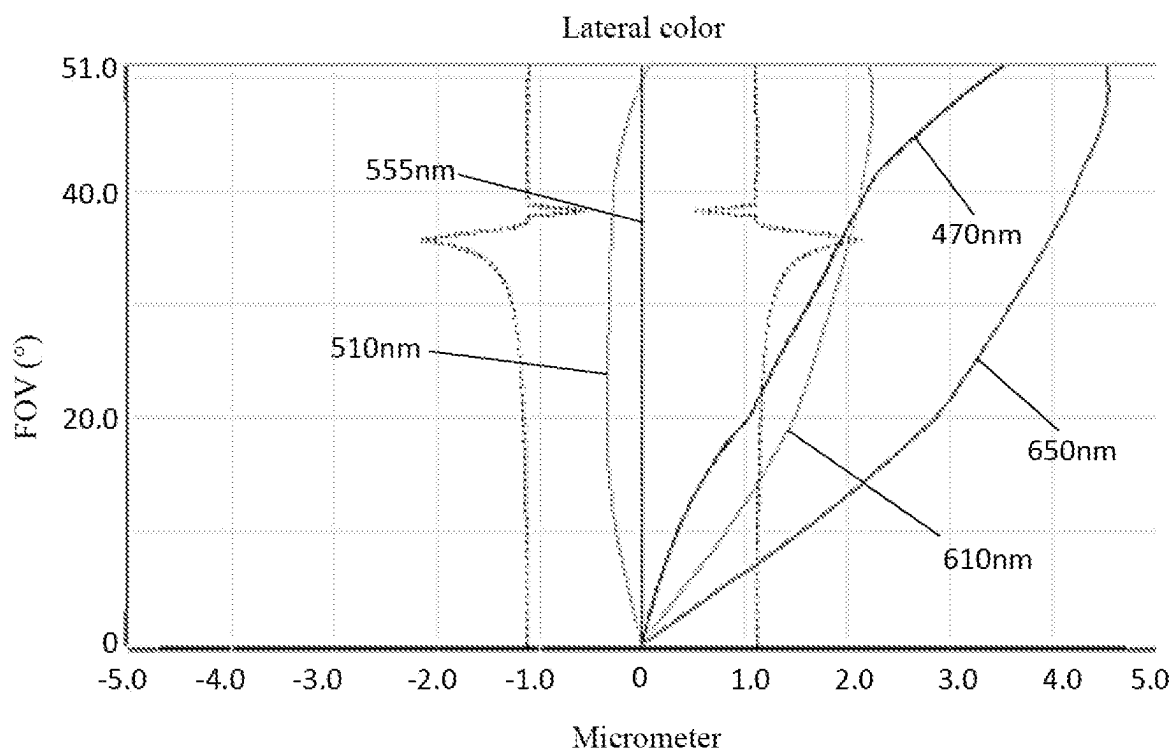


FIG. 23

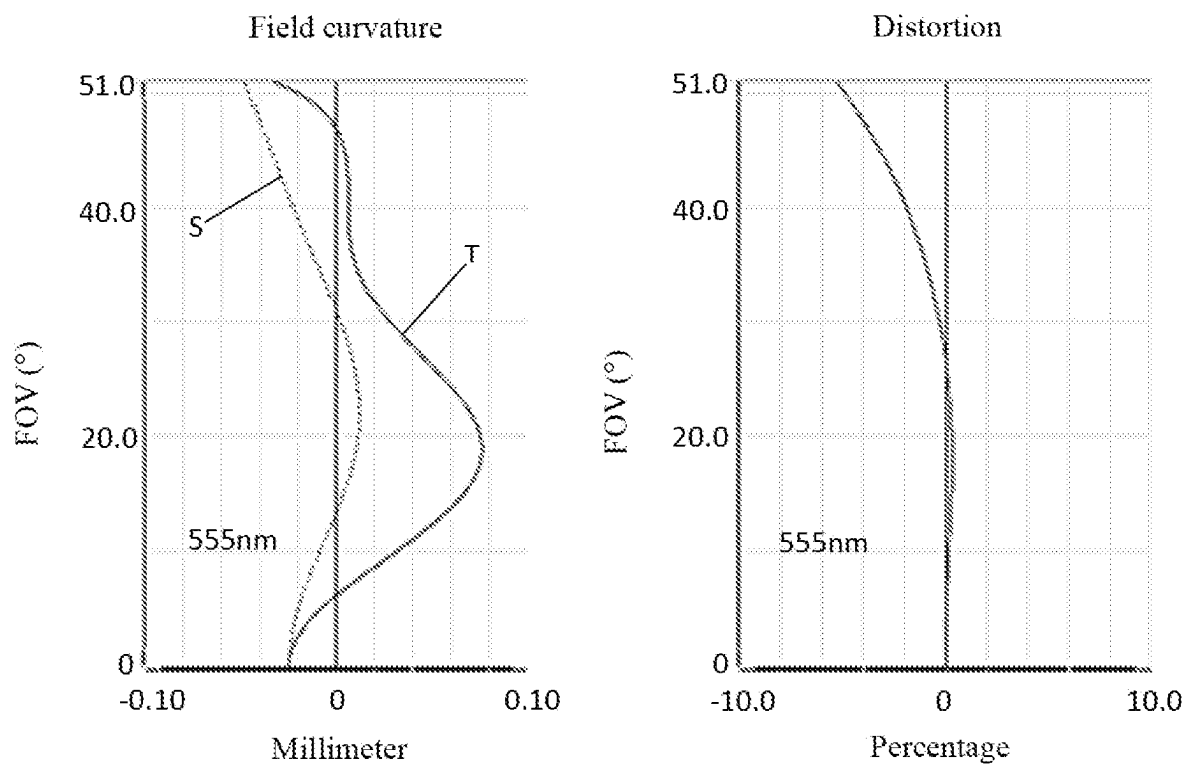


FIG. 24

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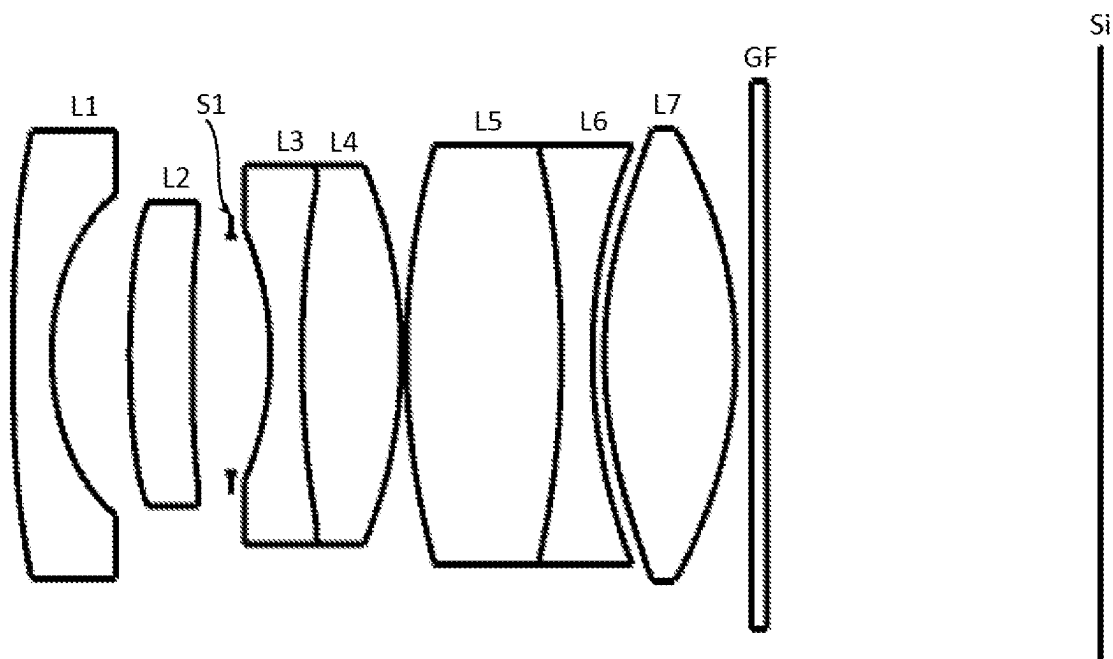


FIG. 25

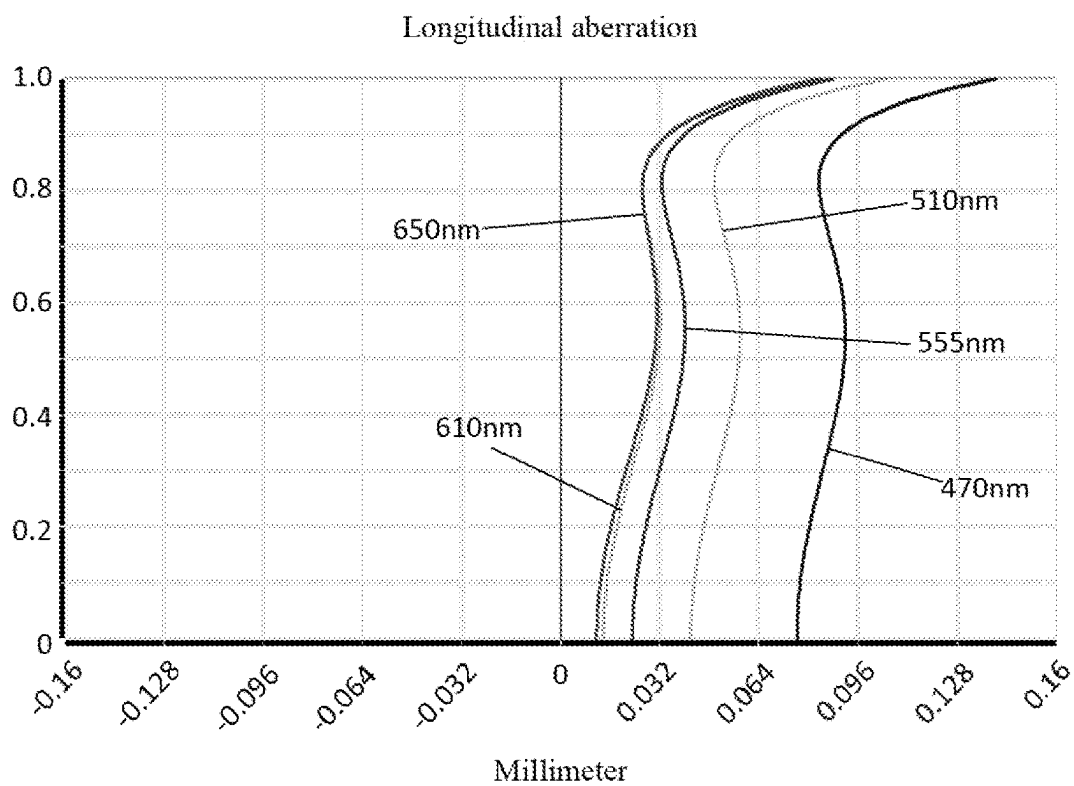


FIG. 26

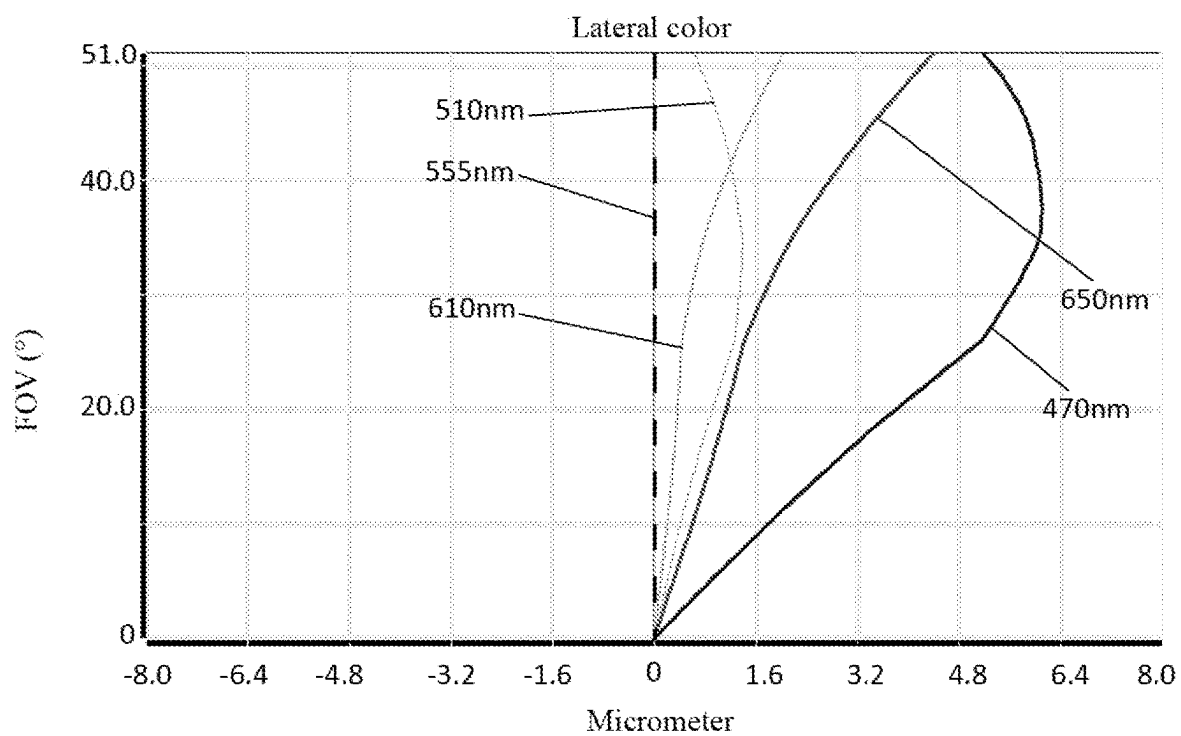


FIG. 27

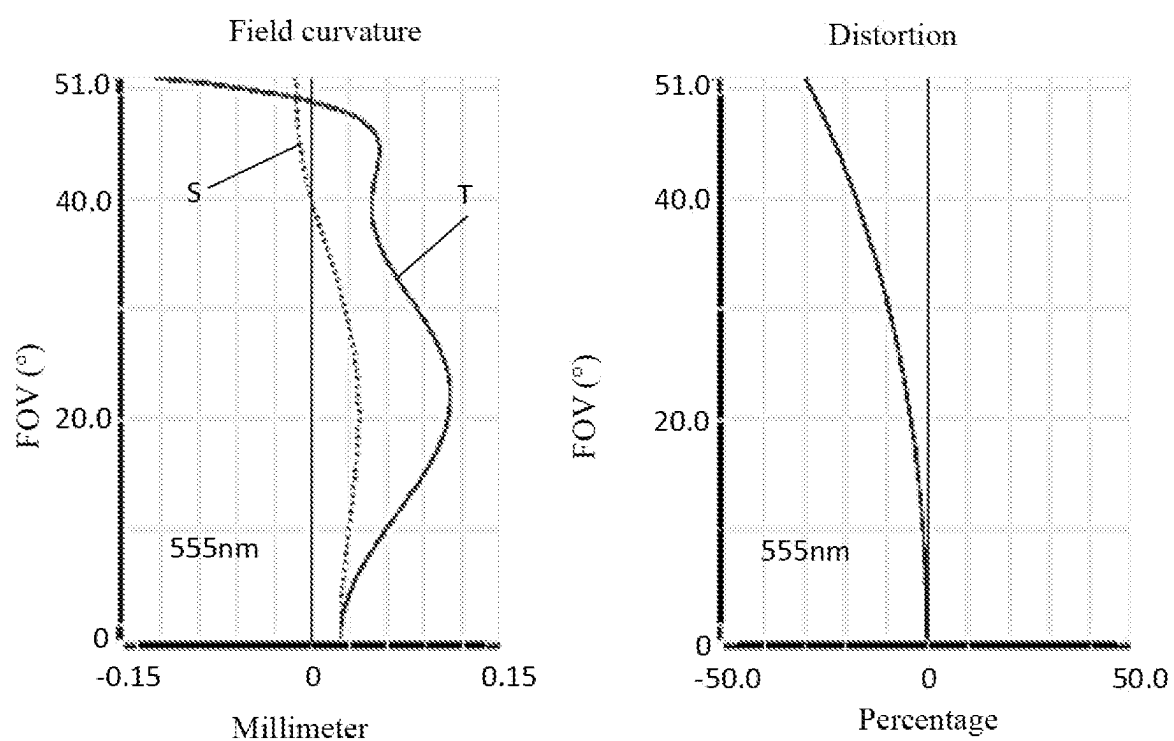


FIG. 28

CAMERA OPTICAL LENS

TECHNICAL FIELD

The present disclosure relates to the technical field of optical lens and, in particular, to a camera optical lens suitable for handheld terminal devices such as smart phones or digital cameras, and imaging devices such as monitors or PC lenses.

BACKGROUND

With the rise of smart phones in recent years, the demand for miniature camera lens is continuously increasing, and since the pixel size of the photo sensors become smaller, along with the current development trend of electronic products that requires better performance such—as thinner and smaller dimensions, miniature camera lenses with good imaging quality therefore have become a mainstream in the market. In order to obtain better imaging quality, a structure with multiple lenses is generally used. Moreover, with development of technology and increase in diversified requirements of users, a camera lens constituted by seven lenses gradually appears in camera lens design, in case that pixel area of the photosensitive device is continuously reduced and requirements on imaging quality is continuously increased. A wide-angle camera optical lens having good optical performance is urgently needed.

SUMMARY

In view of the above problems, the present disclosure provides a camera optical lens, which has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiments of the present disclosure provide a camera optical lens, including from an object side to an image side: a first lens; a second lens; a third lens; a fourth lens; a fifth lens; a sixth lens; and a seventh lens, the camera optical lens satisfies following conditions: $1.30 \leq f_7/f \leq 4.00$; $1.70 \leq n_1$; $2.00 \leq f_3/f \leq 7.00$; and $120.00 \leq (FOV \times f)/IH$, where f denotes a focal length of the camera optical lens; f_7 denotes a focal length of the seventh lens; f_3 denotes a combined focal length of the third lens and the fourth lens; n_1 denotes a refractive index of the first lens; FOV denotes a field of view of the camera optical lens; and IH denotes an image height of the camera optical lens.

As an improvement, the camera optical lens satisfies a following condition: $1.00 \leq f_5/f \leq 3.00$, where f_5 denotes a focal length of the fifth lens.

As an improvement, the camera optical lens satisfies following conditions: $d_{10}=0$; and $0.40 \leq |f_5/f_6| \leq 3.00$, where d_{10} denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens; f_5 denotes a focal length of the fifth lens; and f_6 denotes a focal length of the sixth lens.

As an improvement, the camera optical lens satisfies a following condition: $0.35 \leq BF/TL$, where BF denotes an on-axis distance from an image-side surface of the seventh lens to an image plane of the camera optical lens; and TL denotes an on-axis distance from an object-side surface of the first lens to the image-side surface of the seventh lens.

As an improvement, the camera optical lens satisfies a following condition: $0.01 \leq (d_4+d_0)/TTL \leq 0.10$, where d_4

denotes an on-axis distance from an image-side surface of the second lens to an aperture; d_0 denotes an on-axis distance from the aperture to an object-side surface of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, the camera optical lens satisfies a following condition: $0 < D/IH/FOV \leq 0.10$, where D denotes a light passing diameter of an object-side surface of the first lens corresponding to a maximum field of view of the camera optical lens.

As an improvement, the camera optical lens satisfies a following condition: $0 < TTL/IH/FOV \leq 0.06$, where TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, the camera optical lens satisfies following conditions: $d_6=0$; and $d_{10}=0$, where d_6 denotes an on-axis distance from an image-side surface of the third lens to an object-side surface of the fourth lens; and d_{10} denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens.

As an improvement, an object-side surface of the first lens is convex in a paraxial region, and an image-side surface of the first lens is concave in the paraxial region; the first lens has a negative refractive power; and the camera optical lens satisfies following conditions: $-3.09 \leq f_1/f \leq -0.60$; $0.57 \leq (R_1+R_2)/(R_1-R_2) \leq 1.88$; and $0.02 \leq d_1/TTL \leq 0.05$, where f_1 denotes a focal length of the first lens; R_1 denotes a central curvature radius of the object-side surface of the first lens; R_2 denotes a central curvature radius of the image-side surface of the first lens; d_1 denotes an on-axis thickness of the first lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, the camera optical lens satisfies following conditions: $-10.46 \leq f_2/f \leq 19.43$; $-4.44 \leq (R_3+R_4)/(R_3-R_4) \leq 22.86$; and $0.03 \leq d_3/TTL \leq 0.09$, where f_2 denotes a focal length of the second lens; R_3 denotes a central curvature radius of an object-side surface of the second lens; R_4 denotes a central curvature radius of an image-side surface of the second lens; d_3 denotes an on-axis thickness of the second lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, the camera optical lens satisfies following conditions: $-9.86 \leq f_3/f \leq 1.77$; $-1.65 \leq (R_5+R_6)/(R_5-R_6) \leq 0.53$; and $0.01 \leq d_5/TTL \leq 0.14$, where f_3 denotes a focal length of the third lens; R_5 denotes a central curvature radius of an object-side surface of the third lens; R_6 denotes a central curvature radius of an image-side surface of the third lens; d_5 denotes an on-axis thickness of the third lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, the camera optical lens satisfies following conditions: $-3.27 \leq f_4/f \leq 2.38$; $-0.14 \leq (R_7+R_8)/(R_7-R_8) \leq 1.27$; and $0.01 \leq d_7/TTL \leq 0.23$, where f_4 denotes a focal length of the fourth lens; R_7 denotes a central curvature radius of an object-side surface of the fourth lens; R_8 denotes a central curvature radius of an image-side surface of the fourth lens; d_7 denotes an on-axis thickness of the fourth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, an object-side surface of the fifth lens is convex in a paraxial region and an image-side surface of

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the fifth lens is convex in the paraxial region; the fifth lens has positive refractive power; the camera optical lens satisfies following conditions: $-0.42 \leq (R9+R10)/(R9-R10) \leq 0.84$; and $0.05 \leq d9/TTL \leq 0.22$, where R9 denotes a central curvature radius of an object-side surface of the fifth lens; R10 denotes a central curvature radius of an image-side surface of the fifth lens; d9 denotes an on-axis thickness of the fifth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, an object-side surface of the sixth lens is concave in a paraxial region; the sixth lens has negative refractive power; the camera optical lens satisfies following conditions: $-2.88 \leq f6/f \leq -0.76$; $-2.78 \leq (R11+R12)/(R11-R12) \leq 0.42$; and $0.01 \leq d11/TTL \leq 0.16$, where f6 denotes a focal length of the sixth lens; R11 denotes a central curvature radius of an object-side surface of the sixth lens; R12 denotes a central curvature radius of an image-side surface of the sixth lens; d11 denotes an on-axis thickness of the sixth lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, an object-side surface of the seventh lens is convex in a paraxial region and an image-side surface of the seventh lens is convex in the paraxial region; the seventh lens has positive refractive power; the camera optical lens satisfies following conditions: $-0.50 \leq (R13+R14)/(R13-R14) \leq 0.28$; and $0.05 \leq d13/TTL \leq 0.20$, where R13 denotes a central curvature radius of an object-side surface of the seventh lens; R14 denotes a central curvature radius of an image-side surface of the seventh lens; d13 denotes an on-axis thickness of the seventh lens; and TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

As an improvement, at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens, the sixth lens, and the seventh lens is made of glass material.

The present disclosure has following beneficial effects: the camera optical lens of the present disclosure has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost, which is especially suitable for mobile phone camera lens components and WEB camera lens that are composed of high-pixel CCD, CMOS and other imaging elements, respectively.

BRIEF DESCRIPTION OF DRAWINGS

Many aspects of the exemplary embodiments may be better understood with reference to following drawings. The components in the drawings are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present disclosure. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

FIG. 1 is a structural schematic diagram of a camera optical lens according to Embodiment 1 of the present disclosure;

FIG. 2 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 1;

FIG. 3 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 1;

FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 1;

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FIG. 5 is a structural schematic diagram of a camera optical lens according to Embodiment 2 of the present disclosure;

FIG. 6 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 5;

FIG. 7 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 5;

FIG. 8 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 5;

FIG. 9 is a structural schematic diagram of a camera optical lens according to Embodiment 3 of the present disclosure;

FIG. 10 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 9;

FIG. 11 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 9;

FIG. 12 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 9;

FIG. 13 is a structural schematic diagram of a camera optical lens according to Embodiment 4 of the present disclosure;

FIG. 14 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 13;

FIG. 15 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 13;

FIG. 16 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 13;

FIG. 17 is a structural schematic diagram of a camera optical lens according to Embodiment 5 of the present disclosure;

FIG. 18 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 17;

FIG. 19 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 17;

FIG. 20 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 17;

FIG. 21 is a structural schematic diagram of a camera optical lens according to Embodiment 6 of the present disclosure;

FIG. 22 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 21;

FIG. 23 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 21;

FIG. 24 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 21;

FIG. 25 is a structural schematic diagram of a camera optical lens according to Comparative Example 1 of the present disclosure;

FIG. 26 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 25;

FIG. 27 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 25; and

FIG. 28 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 25.

DESCRIPTION OF EMBODIMENTS

In order to better illustrate the objectives, technical solutions and advantages of the present disclosure, the present disclosure will be described in further detail below with reference to the accompanying drawings and embodiments. It should be understood that the specific embodiments described herein are only used to explain the present disclosure but are not used to limit the present disclosure.

Embodiment 1

Referring to FIG. 1, the present disclosure provides a camera optical lens 10. FIG. 1 is a structural schematic

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diagram of a camera optical lens **10** according to Embodiment 1 of the present disclosure. The camera optical lens **10** includes seven lenses. The camera optical lens **10** includes, from an object side at the left to an image side at the right, a first lens **L1**, a second lens **L2**, an aperture **S1**, a third lens **L3**, a fourth lens **L4**, a fifth lens **L5**, a sixth lens **L6**, and a seventh lens **L7**. An optical element such as an optical filter **GF** may be arranged between the seventh lens **L7** and an image plane **Si**.

In this embodiment, the first lens **L1**, the second lens **L2**, the third lens **L3**, the fourth lens **L4**, the fifth lens **L5**, the sixth lens **L6**, and the seventh lens **L7** are each made of a glass material. In other embodiments, the lenses may also be made of other materials.

A focal length of the camera optical lens **10** is defined as f , and a focal length of the seventh lens **L7** is defined as $f7$. A following condition is satisfied: $1.30 \leq f7/f \leq 4.00$. This condition specifies a ratio of the focal length $f7$ of the seventh lens **L7** to the focal length f of the camera optical lens **10**. Within the range of the above condition, it is beneficial to receive light and ensure the amount of light passing therethrough, so that the camera optical lens **10** has a small CRA (an incidence angle of main light toward the image plane is defined as CRA), and $CRA < 17.5^\circ$.

A refractive index of the first lens **L1** is defined as $n1$. A following condition is satisfied: $1.70 \leq n1$. This condition specifies a range of refractive index of the material used in the first lens **L1**. Within the range of the above condition, it is beneficial to reducing a diameter of front opening and improving imaging quality.

The focal length of the camera optical lens **10** is defined as f , and a combined focal length of the third lens **L3** and the fourth lens **L4** is defined as $f34$. A following condition is satisfied: $2.00 \leq f34/f \leq 7.00$. This condition specifies a ratio of the combined focal length $f34$ of the third lens **L3** and the fourth lens **L4** to the focal length f of the camera optical lens **10**. Within the range of the above condition, the orientation of light between the first lens **L1** and the fourth lens **L4** can be effectively controlled, and the aberration caused by the large-angle light passing through the first lens **L1** can be reduced, while the camera optical lens **10** has a compact structure, which is beneficial to miniaturization.

The focal length of the camera optical lens **10** is defined as f , a field of view of the camera optical lens **10** is defined as FOV, an image height of the camera optical lens **10** is defined as IH . A following condition is satisfied: $120.00 \leq (FOV \times f)/IH$. This condition specifies a ratio of product of the field of view FOV of the camera optical lens **10** and the focal length f of the camera optical lens **10** to the image height IH of the camera optical lens **10**. Within the range of the above condition, the camera optical lens **10** can be designed with both long focal length and large field of view.

The focal length of the camera optical lens **10** is defined as f , and the focal length of the fifth lens **L5** is defined as $f5$. A following condition is satisfied: $1.00 \leq f5/f \leq 3.00$. This condition specifies a ratio of the focal length $f5$ of the fifth lens **L5** to the focal length f of the camera optical lens **10**. By controlling the focal length reasonably, within the range of the above condition, it is beneficial to controlling temperature drift and thus has good temperature performance.

The focal length of the fifth lens **L5** is defined as $f5$, and a focal length of the sixth lens **L6** is defined as $f6$. An on-axis distance from an image-side surface of the fifth lens **L5** to an object-side surface of the sixth lens **L6** is defined as $d10$. Following conditions are satisfied: $d10 > 0$; and $0.40 \leq f5/f6 \leq 3.00$. A ratio of the focal length $f5$ of the lens **L5** to the focal length $f6$ of the lens **L6** is specified. Within the range

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of the above condition, the absolute values of focal length of two lens of glued lenses are close to each other, it is beneficial to smooth transition of light, improving imaging quality, and causing the chromatic aberration to be $|LC| \leq 10 \mu m$.

An on-axis distance from the image-side surface of the seventh lens **L7** to the image plane is defined as BF, and an on-axis distance from the image-side surface of the first lens **L1** to the image-side surface of the seventh lens **L7** is TL. A following condition is satisfied: $0.35 \leq BF/TL$. This condition specifies a ratio of the on-axis distance BF from the image-side surface of the seventh lens **L7** to the image plane to the on-axis distance from the image-side surface of the first lens **L1** to the image-side surface of the seventh lens **L7**. On the basis of miniaturization, the back focal length is long, which is beneficial to assembling the modules. A length TL of lens group TL is short, the structure is compact, which is beneficial to reducing the sensitivity of the lens to MTF, improving the production yield and reducing the production cost.

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, an on-axis distance from an image-side surface of the second lens **L2** to an aperture **S1** is defined as $d4$, and an on-axis distance from the aperture **S1** to an object-side surface of the third lens **L3** is defined $d0$. A following condition is satisfied: $0.01 \leq (d4 + d0)/TTL \leq 0.10$. This condition specifies a ratio of the on-axis distance from the second lens **L2** to the third lens **L3** to the total length TTL of the camera optical lens. Within the range of the above condition, the distance between two lenses adjacent to the aperture **S1** is increased, which is beneficial to improving the imaging quality.

The field of view of the camera optical lens **10** is defined as FOV, and the image height of the camera optical lens **10** is defined as IH . A light passing diameter of an object-side surface of the first lens **L1** corresponding to a maximum field of view of the camera optical lens **10** is defined as D . A following condition is satisfied: $0 < D/IH/FOV \leq 0.10$. Within the range of the above condition, it is beneficial to achieving miniaturization and small diameter of front opening.

The field of view of the camera optical lens **10** is defined as FOV, the image height of the camera optical lens **10** is defined as IH , and the optical total length of the camera optical lens is TTL. A following condition is satisfied: $0 < TTL/IH/FOV \leq 0.06$. Within the range of the above condition, it is beneficial to achieving miniaturization and small diameter of front opening.

An on-axis distance from an image-side surface of the third lens **L3** to an object-side surface of the fourth lens **L4** is defined as $d6$, and an on-axis distance from an image-side surface of the fifth lens **L5** to an object-side surface of the sixth lens **L6** is defined as $d10$. A following condition is satisfied: $d6 = 0$; $d10 = 0$. The use of two sets of glued lenses is not only beneficial to correcting aberration to achieve high resolution, but also to reducing the problems of tolerance sensitivity of lens units such as tilt/core shift caused in the assembling process.

In this embodiment, the first lens **L1** has a negative refractive power. An object-side surface of the first lens **L1** is convex in a paraxial region, and an image-side surface of the first lens **L1** is concave in the paraxial region. In other embodiments, the first lens **L1** may also have a positive refractive power, and the object-side surface and image-side surface of the first lens **L1** may also have other concave or convex distribution in the paraxial region.

The focal length of the camera optical lens **10** is defined as f , and the focal length of the first lens **L1** is defined as f_1 . A following condition is satisfied: $-3.09 \leq f_1/f \leq -0.60$. This condition specifies a ratio of the focal length f_1 of the first lens **L1** to the focal length f of the camera optical lens **10**. Within the range of the above condition, the first lens has an appropriate negative refractive power, which is beneficial to reducing the aberration of the system while developing ultra-thinness and wide angle of the lens. Optionally, a following condition is satisfied: $-1.93 \leq f_1/f \leq -0.75$.

A central curvature radius of the object-side surface of the first lens **L1** is defined as R_1 , and a central curvature radius of the image-side surface of the first lens **L1** is R_2 . A following condition is satisfied: $0.57 \leq (R_1 + R_2)/(R_1 - R_2) \leq 1.88$. The shape of the first lens **L1** is controlled reasonably, so that the first lens **L1** can effectively correct the spherical aberration of the camera optical lens **10**. Optionally, a following condition is satisfied: $0.91 \leq (R_1 + R_2)/(R_1 - R_2) \leq 1.51$.

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the first lens **L1** is defined as d_1 . A following condition is satisfied: $0.02 \leq d_1/TTL \leq 0.05$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.03 \leq d_1/TTL \leq 0.04$.

In this embodiment, the second lens **L2** has a positive refractive power. An object-side surface of the second lens **L2** is concave in a paraxial region, and an image-side surface of the second lens **L2** is convex in the paraxial region. In other embodiments, the second lens **L2** may also have a negative refractive power, and the object-side surface and image-side surface of the second lens **L2** may also have other concave or convex distribution in the paraxial region.

The focal length of the camera optical lens **10** is defined as f , and the focal length of the second lens **L2** is defined as f_2 . A following condition is satisfied: $-10.46 \leq f_2/f \leq 19.43$. By controlling the optical focal length of the second lens **L2** within a reasonable range, it is beneficial to correct the aberration of the camera optical lens **10**. Optionally, a following condition is satisfied: $-6.54 \leq f_2/f \leq 15.54$.

A central curvature radius of the object-side surface of the second lens **L2** is defined as R_3 , and a central curvature radius of the image-side surface of the second lens **L2** is defined as R_4 . A following condition is satisfied: $-4.44 \leq (R_3 + R_4)/(R_3 - R_4) \leq 22.86$, which specifies the shape of the second lens **L2**. Within the range of the above condition, with the development of the camera optical lens **10** to ultra-thinness and wide angle, it is beneficial to correcting the on-axis chromatic aberration. Optionally, a following condition is satisfied:

$$-2.78 \leq (R_3 + R_4)/(R_3 - R_4) \leq 18.29.$$

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and the on-axis thickness of the second lens **L2** is defined as d_3 . A following condition is satisfied: $0.03 \leq d_3/TTL \leq 0.09$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.04 \leq d_3/TTL \leq 0.07$.

In this embodiment, the third lens **L3** has a negative refractive power. An object-side surface of the third lens **L3**

is concave in a paraxial region, and an image-side surface of the third lens **L3** is concave in the paraxial region. In other embodiments, the third lens **L3** may also have a positive refractive power, and the object-side surface and image-side surface of the third lens **L3** may also have other concave or convex distribution in the paraxial region.

The focal length of the camera optical lens **10** is defined as f , and the focal length of the third lens **L3** is defined as f_3 . A following condition is satisfied: $-9.86 \leq f_3/f \leq 1.77$. The optical focal length is controlled reasonably, so that the camera optical lens **10** has better imaging quality and lower sensitivity. Optionally, a following condition is satisfied:

$$-6.17 \leq f_3/f \leq 1.42.$$

A central curvature radius of the object-side surface of the third lens **L3** is defined as R_5 , and a central curvature radius of the image-side surface of the third lens **L3** is defined as R_6 . A following condition is satisfied: $-1.65 \leq (R_5 + R_6)/(R_5 - R_6) \leq 0.53$, which specifies the shape of the third lens **L3**. Within the range of the above condition, it is beneficial to molding the third lens **L3**, which can alleviate the deflection degree of light passing through the lens and effectively reduce the aberration. Optionally, a following condition is satisfied:

$$-1.03 \leq (R_5 + R_6)/(R_5 - R_6) \leq 0.43.$$

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the third lens **L3** is defined as d_5 . A following condition is satisfied: $0.01 \leq d_5/TTL \leq 0.14$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.02 \leq d_5/TTL \leq 0.11$.

In this embodiment, the fourth lens **L4** has a positive refractive power. The object-side surface of the fourth lens **L4** is convex in the paraxial region, and the image side of the fourth lens **L4** is convex in the paraxial region. In other embodiments, the fourth lens **L4** may also have a negative refractive power, and the object-side surface and image-side surface of the fourth lens **L4** may also have other concave or convex distribution in the paraxial region.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the fourth lens **L4** is defined as f_4 . A following condition is satisfied: $-3.27 \leq f_4/f \leq 2.38$, which specifies a ratio of the focal length f_4 of the fourth lens **L4** to the focal length f of the camera optical lens **10**. The optical focal length is controlled reasonably, so that the camera optical lens **10** has better imaging quality and lower sensitivity. Optionally, a following condition is satisfied: $-2.05 \leq f_4/f \leq 1.90$.

A central curvature radius of the object-side surface of the fourth lens **L4** is defined as R_7 , and a central curvature radius of the image-side surface of the fourth lens **L4** is defined as R_8 . A following condition is satisfied: $-0.14 \leq (R_7 + R_8)/(R_7 - R_8) \leq 1.27$, which specifies the shape of **L4** of the fourth lens. Within the range of the above condition, with the development of ultra-thinness and wide angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied: $-0.09 \leq (R_7 + R_8)/(R_7 - R_8) \leq 1.01$.

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the fourth lens **L4** is defined as d7. A following condition is satisfied: $0.01 \leq d7/TTL \leq 0.23$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.02 \leq d7/TTL \leq 0.19$.

In this embodiment, the fifth lens **L5** has a positive refractive power. The object-side surface of the fifth lens **L5** is convex in the paraxial region, and the image side of the fifth lens **L5** is convex in the paraxial region. In other embodiments, the fifth lens **L5** may also have a negative refractive power, and the object-side surface and image-side surface of the fifth lens **L5** may also have other concave or convex distribution in the paraxial region.

A central curvature radius of the object-side surface of the fifth lens **L5** is defined as R9, and a central curvature radius of the image-side surface of the fifth lens **L5** is defined as R10. A following condition is satisfied: $-0.42 \leq (R9+R10)/(R9-R10) \leq 0.84$, which specifies the shape of the fifth lens **L5**. Within the range of the above condition, with the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied: $-0.26 \leq (R9+R10)/(R9-R10) \leq 0.67$.

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the fifth lens **L5** is defined as d9. A following condition is satisfied: $0.05 \leq d9/TTL \leq 0.22$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.09 \leq d9/TTL \leq 0.17$.

In this embodiment, the sixth lens **L6** has a negative refractive power. The object-side surface of the sixth lens **L6** is concave in the paraxial region, and the image side of the sixth lens **L6** is concave in the paraxial region. In other embodiments, the sixth lens **L6** may also have a positive refractive power, and the object-side surface and image-side surface of the sixth lens **L6** may also have other concave or convex distribution in the paraxial region.

The focal length of the optical lens **10** is defined as f, and the focal length of the sixth lens **L6** is defined as f6. A following condition is satisfied: $-2.88 \leq f6/f \leq -0.76$. The optical focal length is controlled reasonably, so that the camera optical lens **10** has better imaging quality and lower sensitivity. Optionally, a following condition is satisfied: $-1.80 \leq f6/f \leq -0.95$.

A central curvature radius of the object-side surface of the sixth lens **L6** is defined as R11 and a central curvature radius of the image-side surface of the sixth lens **L6** is defined as R12. A following condition is satisfied: $-2.78 \leq (R11+R12)/(R11-R12) \leq 0.42$, which specifies the shape of the sixth lens **L6**. Within the range of the above condition, with the development of ultra-thinness and wide angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied: $-1.74 \leq (R11+R12)/(R11-R12) \leq 0.33$.

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the sixth lens **L6** is defined as d11. A following condition is satisfied: $0.01 \leq d11/TTL \leq 0.16$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.02 \leq d11/TTL \leq 0.12$.

In this embodiment, the seventh lens **L7** has a positive refractive power. The object-side surface of the seventh lens **L7** is convex in the paraxial region, and the image side of the seventh lens **L7** is convex in the paraxial region. In other embodiments, the seventh lens **L7** may also have a negative refractive power, and the object-side surface and image-side surface of the seventh lens **L7** may also have other concave or convex distribution in the paraxial region.

A central curvature radius of the object-side surface of the seventh lens **L7** is defined as R13, and a central curvature radius of the image-side surface of the seventh lens **L7** is defined as R14. A following condition is satisfied: $-0.50 \leq (R13+R14)/(R13-R14) \leq 0.28$, which specifies the shape of the seventh lens **L7**. Within the range of the above condition with the development of ultra-thinness and wide-angle, it is beneficial to correcting aberration of off-axis angle. Optionally, a following condition is satisfied:

$$-0.31 \leq (R13 + R14)/(R13 - R14) \leq 0.23.$$

A total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL, and an on-axis thickness of the seventh lens **L7** is defined as d13. A following condition is satisfied: $0.05 \leq d13/TTL \leq 0.20$. Within the range of the above condition, it is beneficial to achieving ultra-thinness. Optionally, a following condition is satisfied: $0.08 \leq d13/TTL \leq 0.16$.

In this embodiment, the F number of the camera optical lens **10** is defined as FNO. A following condition is satisfied: $FNO \leq 1.75$, so that it is beneficial to achieving a large aperture. Optionally, a following condition is satisfied: $FNO \leq 1.72$.

In this embodiment, the F number of the camera optical lens **10** is defined as FNO, and a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens **10** along an optic axis is defined as TTL. A following condition is satisfied: $TTL/FNO \leq 5.00$, achieving miniaturization and small diameter of front opening.

When the above conditions are satisfied, the camera optical lens **10** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost. According to properties of the camera optical lens **10**, the camera optical lens **10** is especially suitable for mobile phone camera lens components and WEB camera lens that are composed of high-pixel CCD, CMOS and other imaging elements, respectively.

The camera optical lens **10** of the present disclosure will be described below with examples. The symbols recorded in each example will be described as follows. The focal length, on-axis distance, central curvature radius, on-axis thickness, inflection point, and arrest point are each in unit of millimeter (mm).

TTL denotes a total optical length (on-axis distance from the object-side surface of the first lens **L1** to the image plane **Si**), with a unit of millimeter (mm).

F number FNO denotes a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.

Optionally, the object-side surface and/or the image-side surface of the lens may be provided with inflection points and/or arrest points in order to meet high-quality imaging

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requirements. The description below may be referred to in specific embodiments as follows.

Design data of the camera optical lens **10** according to Embodiment 1 of the present disclosure are shown in Tables 1 and 2.

TABLE 1

	R	d	nd	vd
R1	58.524	d1=	1.000	49.29
R2	6.614	d2=	1.998	
R3	-13.439	d3=	1.611	41.00
R4	-11.783	d4=	0.500	
S1	∞	d0=	0.661	
R5	-7.342	d5=	0.806	60.37
R6	16.942	d6=	0.000	
R7	16.942	d7=	4.500	32.32
R8	-9.598	d8=	0.134	
R9	17.864	d9=	3.288	54.57
R10	-12.991	d10=	0.000	
R11	-12.991	d11=	0.806	18.90
R12	19.451	d12=	0.600	
R13	12.362	d13=	4.000	61.16
R14	-10.000	d14=	0.403	
R15	∞	d15=	0.400	64.21
R16	∞	d16=	9.297	

The reference signs are explained as follows.

S1: aperture;

R: central curvature radius of an optical surface;

R1: central curvature radius of the object-side surface of the first lens **L1**;

R2: central curvature radius of the image-side surface of the first lens **L1**;

R3: central curvature radius of the object-side surface of the second lens **L2**;

R4: central curvature radius of the image-side surface of the second lens **L2**;

R5: central curvature radius of the object-side surface of the third lens **L3**;

R6: central curvature radius of the image-side surface of the third lens **L3**;

R7: central curvature radius of the object-side surface of the fourth lens **L4**;

R8: central curvature radius of the image-side surface of the fourth lens **L4**;

R9: central curvature radius of the object-side surface of the fifth lens **L5**;

R10: central curvature radius of the image-side surface of the fifth lens **L5**;

R11: central curvature radius of the object-side surface of the sixth lens **L6**;

R12: central curvature radius of the image-side surface of the sixth lens **L6**;

R13: central curvature radius of the object-side surface of the seventh lens **L7**;

R14: central curvature radius of the image-side surface of the seventh lens **L7**;

R15: central curvature radius of the object-side surface of the optical filter GF;

R16: central curvature radius of the image-side surface of the optical filter GF;

d: on-axis thickness of a lens and an on-axis distance between lenses;

d0: on-axis distance from the aperture S1 to the object-side surface of the third lens **L3**;

d1: on-axis thickness of the first lens **L1**;

d2: on-axis distance from the image-side surface of the first lens **L1** to the object-side surface of the second lens **L2**;

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d3: on-axis thickness of the second lens **L2**;

d4: on-axis distance from the image-side surface of the second lens **L2** to the aperture S1;

d5: on-axis thickness of the third lens **L3**;

d6: on-axis distance from the image-side surface of the third lens **L3** to the object-side surface of the fourth lens **L4**;

d7: on-axis thickness of the fourth lens **L4**;

d8: on-axis distance from the image-side surface of the fourth lens **L4** to the object-side surface of the fifth lens **L5**;

d9: on-axis thickness of the fifth lens **L5**;

d10: on-axis distance from the image-side surface of the fifth lens **L5** to the object-side surface of the sixth lens **L6**;

d11: on-axis thickness of the sixth lens **L6**;

d12: on-axis distance from the image-side surface of the sixth lens **L6** to the object-side surface of the seventh lens **L7**;

d13: on-axis thickness of the seventh lens **L7**;

d14: on-axis distance from the image-side surface of the seventh lens **L7** to the object-side surface of the optical filter GF;

d15: on-axis thickness of the optical filter GF;

d16: on-axis distance from the image-side surface of the optical filter GF to the image plane S_i;

nd: refractive index of a d-line (the d-line is green light with a wavelength of 555 nm);

nd1: refractive index of the d-line of the first lens **L1**;

nd2: refractive index of the d-line of the second lens **L2**;

nd3: refractive index of the d-line of the third lens **L3**;

nd4: refractive index of the d-line of the fourth lens **L4**;

nd5: refractive index of the d-line of the fifth lens **L5**;

nd6: refractive index of the d-line of the sixth lens **L6**;

nd7: refractive index of the d-line of the seventh lens **L7**;

ndg: refractive index of the d-line of the optical filter GF;

vd: Abbe number;

v1: Abbe number of the first lens **L1**;

v2: Abbe number of the second lens **L2**;

v3: Abbe number of the third lens **L3**;

v4: Abbe number of the fourth lens **L4**;

v5: Abbe number of the fifth lens **L5**;

v6: Abbe number of the sixth lens **L6**;

v7: Abbe number of the seventh lens **L7**;

vg: Abbe number of the optical filter GF.

Table 2 shows aspheric surface data of the first lens **L1**, the second lens **L2** and the seventh lens **L7** in the camera optical lens **10** according to Embodiment 1 of the present disclosure.

It should be noted that that each lens surface of the third lens **L3**, the fourth lens **L4**, the fifth lens **L5** and the sixth lens **L6** in the camera optical lens **10** in Embodiment 1 is a spherical surface.

TABLE 2

	Conic coefficient	Aspheric coefficient				
	k	A4	A6	A8	A10	A12
R1	1.2702E+02	1.9741E-03	-1.3814E-04	6.1484E-06	-1.4288E-07	9.0215E-10
R2	5.4438E-01	2.9063E-03	-6.3457E-05	-6.4599E-06	1.2949E-06	-2.3331E-08
R3	3.3777E+00	-1.5381E-04	4.2920E-06	-4.6369E-06	9.6074E-07	-4.2277E-08
R4	-1.2799E-01	-8.3240E-05	9.2990E-06	-3.2090E-06	6.4871E-07	-3.2714E-08
R13	-2.4460E+00	-5.1900E-06	1.9260E-07	3.5846E-08	-2.7044E-10	6.0425E-12
R14	-2.1602E+00	1.3547E-05	9.4447E-09	4.9161E-08	-5.3020E-10	1.3417E-11

For convenience, the aspheric surfaces of the first lens **L1**, the second lens **L2**, and the seventh lens **L7** use the aspheric surfaces shown in formula (1) below. However, the present disclosure is not limited to the polynomial form of aspheric surfaces represented by formula (1).

$$z = (cr^2) \left\{ 1 + [1 + (k+1)(c^2r^2)]^{1/2} \right\} + A4r^4 + A6r^6 + A8r^8 + A10r^{10} + A12r^{12} \quad (1)$$

Here, k denotes a conic coefficient, and A4, A6, A8, A10, and A12 each denote an aspheric coefficient, respectively, c denotes a curvature at a center of an optical surface, r denotes a vertical distance between a point on a curve of the aspheric surface and the optical axis, and Z denotes an aspheric depth (i.e., a vertical distance between a point of the aspheric surface from which a distance to the optical axis is r, and a tangent plane tangent to a vertex on an aspheric optical axis).

Design data of the inflection point and the arrest point of each lens in the camera optical lens **10** according to Embodiment 1 of the present disclosure are shown in Tables 3 and 4. Here, P1R1 and P1R2 denote the object side surface and image side surface of the first lens **L1**, respectively. P2R1 and P2R2 denote the object side surface and image side surface of the second lens **L2**, respectively. P3R1 and P3R2 denote the object side surface and image side surface of the third lens **L3**, respectively. P4R1 and P4R2 denote the object side surface and image side surface of the fourth lens **L4**, respectively. P5R1 and P5R2 denote the object side surface and image side surface of the fifth lens **L5**, respectively. P6R1 and P6R2 denote the object side surface and image side surface of the sixth lens **L6**, respectively. P7R1 and P7R2 denote the object side surface and image side surface of the seventh lens **L7**, respectively. Data in an “inflection point” column are a vertical distance from an inflexion point provided on a surface of each lens to the optical axis of the camera optical lens **10**. Data in an “arrest point” column are a vertical distance from an arrest point provided on the surface of each lens to the optical axis of the camera optical lens **10**.

TABLE 3

	Number of inflection points	Inflection point 1
P1R1	0	/
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/

TABLE 3-continued

	Number of inflection points	Inflection point 1
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	1	5.285

TABLE 4

	Number of arrest points	Arrest point1
P1R1	0	/
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	0	/

FIG. 2 and FIG. 3 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens **10** after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens **10** according to Embodiment 1, respectively. FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens **10** after light having a wavelength of 555 nm passes through the camera optical lens **10** according to Embodiment 1. A field curvature S in FIG. 4 is a field curvature in a sagittal direction, and T is a field curvature in a meridian direction.

Table 29 below shows numerical values corresponding to various numerical values in Embodiments 1, 2, 3, 4 and 5, and parameters specified in the conditions. As shown in Table 29, Embodiment 1 satisfies various conditions.

In this embodiment, the camera optical lens **10** has an entrance pupil diameter ENPD of 3.950 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 102.40° in a diagonal direction. The camera optical lens **10** has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 2

FIG. 5 is a structural schematic diagram of a camera optical lens **20** according to Embodiment 2 of the present

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disclosure. Embodiment 2 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

In this embodiment, the object-side surface of the second lens L2 is convex in the paraxial region, and the image-side surface of the second lens L2 is concave in the paraxial region.

Table 5 and Table 6 show the design data of the camera optical lens 20 of the Embodiment 2 of the present disclosure.

TABLE 5

	R	d	nd	vd
R1	50.000	d1=	1.000	nd1 1.7725 v1 49.61
R2	4.953	d2=	2.000	
R3	17.259	d3=	1.611	nd2 1.7433 v2 49.40
R4	50.000	d4=	1.000	
S1	∞	d0=	1.000	
R5	-7.519	d5=	0.806	nd3 1.5174 v3 52.19
R6	78.254	d6=	0.000	
R7	78.254	d7=	2.551	nd4 1.9537 v4 32.32
R8	-8.504	d8=	0.133	
R9	14.042	d9=	3.974	nd5 1.6921 v5 54.57
R10	-21.404	d10=	0.000	
R11	-21.404	d11=	0.806	nd6 1.9229 v6 18.90
R12	12.093	d12=	0.300	
R13	10.516	d13=	3.370	nd7 1.6180 v7 63.41
R14	-9.803	d14=	0.403	
R15	∞	d15=	0.400	ndg 1.5168 vg 64.21
R16	∞	d16=	8.826	

Table 6 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 20 according to Embodiment 2 of the present disclosure.

TABLE 6

	Conic coefficient	Aspheric coefficient				
	k	A4	A6	A8	A10	A12
R1	7.3818E+01	1.5470E-04	2.3928E-06	-9.0391E-08	-6.0759E-09	-5.6886E-11
R2	2.7310E-01	3.5301E-04	4.9216E-05	1.1979E-06	1.5973E-07	4.0319E-08
R3	-4.1838E+00	1.3787E-04	2.0759E-05	3.0865E-06	2.4227E-07	6.6851E-09
R4	6.4431E+01	4.9935E-05	1.3584E-05	1.3183E-06	1.2994E-07	1.0727E-08
R13	-2.1026E+00	-2.8727E-06	4.9172E-07	2.7468E-08	1.3838E-09	3.1063E-11
R14	-1.9136E+00	-9.6042E-06	-1.5748E-07	8.6307E-09	9.6703E-10	9.0303E-11

Design data of the inflection point and the arrest point of each lens in the camera optical lens 20 according to Embodiment 2 of the present disclosure are shown in Tables 7 and 8.

TABLE 7

	Number of inflection points	Inflection point 1
P1R1	1	4.425
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/

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TABLE 7-continued

	Number of inflection points	Inflection point 1
P7R1	0	/
P7R2	1	4.665

TABLE 8

	Number of arrest points	Arrest point1
P1R1	0	/
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	0	/

FIG. 6 and FIG. 7 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 20 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 20 according to Embodiment 2, respectively. FIG. 8 is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens 20 according to Embodiment 2. The field curvature S in FIG. 8 is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

As shown in Table 29, Embodiment 2 satisfies various conditions.

In this embodiment, the camera optical lens 20 has an entrance pupil diameter ENPD of 3.930 mm, a full-field image height IH of 5.145 mm, and a field of view FOV 102.10° in a diagonal direction. The camera optical lens 20 has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 3

FIG. 9 is a structural schematic diagram of a camera optical lens 30 according to Embodiment 3 of the present disclosure. Embodiment 3 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

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In this embodiment, the object-side surface of the third lens L3 is convex in the paraxial region, and the image-side surface of the third lens L3 is convex in the paraxial region. The object-side surface of the fourth lens L4 is concave in the paraxial region, and the image-side surface of the fourth lens L4 is concave in the paraxial region. The image-side surface of the sixth lens L6 is convex in the paraxial region. The third lens L3 has a positive refractive power, and the fourth lens L4 has a negative refractive power.

Design data of the camera optical lens 30 of Embodiment 3 of the present disclosure are shown in Tables 9 and 10.

TABLE 9

	R	d	nd	vd
R1	50.000	d1=	1.000	nd1
R2	4.456	d2=	2.940	nd1
R3	-30.000	d3=	1.611	nd2
R4	-21.000	d4=	0.600	nd2
S1	∞	d0=	0.000	
R5	21.146	d5=	2.551	nd3
R6	-10.070	d6=	0.000	nd3
R7	-10.070	d7=	0.800	nd4
R8	11.606	d8=	0.633	nd4
R9	10.107	d9=	3.974	nd5
R10	-6.878	d10=	0.000	nd5
R11	-6.878	d11=	0.806	nd6
R12	-77.195	d12=	0.205	nd6
R13	14.278	d13=	3.359	nd7
R14	-23.865	d14=	0.403	nd7
R15	∞	d15=	0.400	ndg
R16	∞	d16=	8.248	ndg

Table 10 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 30 of the Embodiment 3 of the present disclosure.

TABLE 10

	Conic coefficient	Aspheric coefficient				
	k	A4	A6	A8	A10	A12
R1	1.0563E+02	-3.6645E-04	-3.2648E-05	9.9226E-07	2.0032E-08	-1.3970E-09
R2	5.7717E-02	6.9303E-04	-5.4109E-05	-7.7723E-08	1.2845E-07	1.3919E-09
R3	-8.3399E+01	7.2850E-04	-9.8556E-06	1.2989E-06	7.2341E-08	1.9118E-09
R4	-1.4483E+01	1.5478E-04	-1.0031E-05	-8.0541E-07	1.1500E-07	-2.0905E-10
R13	3.8745E+00	2.3772E-04	2.6665E-05	-1.6177E-06	5.9282E-08	-7.5860E-10
R14	-1.1735E+01	9.9744E-04	1.0816E-05	1.7072E-06	-8.8881E-08	2.6969E-09

Design data of the inflection point and the arrest point of each lens in the camera optical lens 30 according to Embodiment 3 of the present disclosure are shown in Tables 11 and 12.

TABLE 11

	Number of inflection points	Inflection point 1	Inflection point 2
P1R1	2	1.905	4.455
P1R2	0	/	/
P2R1	1	1.665	/
P2R2	1	3.075	/
P3R1	0	/	/
P3R2	0	/	/
P4R1	0	/	/
P4R2	0	/	/
P5R1	0	/	/
P5R2	0	/	/
P6R1	0	/	/
P6R2	0	/	/

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TABLE 11-continued

	Number of inflection points	Inflection point 1	Inflection point 2
P7R1	0	/	/
P7R2	1	1.695	/

TABLE 12

	Number of arrest points	Arrest point1
P1R1	1	3.195
P1R2	0	/
P2R1	1	2.735
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	1	2.795

FIG. 10 and FIG. 11 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 30 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 30 according to Embodiment 3, respectively. FIG. 12 is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens 30 according to Embodiment 3. The field curvature S in FIG. 12 is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens 30 of this embodiment satisfies the above conditions.

In this embodiment, the camera optical lens 30 has an entrance pupil diameter ENPD of 3.915 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 102.60° in a diagonal direction. The camera optical lens 30 has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 4

FIG. 13 is a structural schematic diagram of a camera optical lens 40 according to Embodiment 4 of the present disclosure. Embodiment 4 is basically the same as Embodi-

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ment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

In this embodiment, the object-side surface of the second lens L3 is convex in the paraxial region, and the image-side surface of the sixth lens L6 is convex in the paraxial region.

Design data of the camera optical lens 40 of Embodiment 4 of the present disclosure are shown in Tables 13 and 14.

TABLE 13

	R	d	nd	vd
R1	68.688	d1=	1.000 nd1	1.8839 v1
R2	4.605	d2=	3.426	
R3	50.000	d3=	1.700 nd2	1.8839 v2
R4	-24.531	d4=	0.100	
S1	∞	d0=	0.367	
R5	-22.927	d5=	0.806 nd3	1.6204 v3
R6	120.000	d6=	0.000	
R7	120.000	d7=	4.500 nd4	1.9537 v4
R8	-10.132	d8=	0.134	
R9	11.358	d9=	4.000 nd5	1.4970 v5
R10	-6.931	d10=	0.000	
R11	-6.931	d11=	0.806 nd6	1.9229 v6
R12	-42.255	d12=	0.400	
R13	28.200	d13=	3.000 nd7	1.5891 v7
R14	-29.094	d14=	0.403	
R15	∞	d15=	0.400 ndg	1.5168 vg
R16	∞	d16=	7.974	

Table 14 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 40 according to Embodiment 4 of the present disclosure.

TABLE 14

	Conic coefficient		Aspheric coefficient				
	k	A4	A6	A8	A10	A12	
R1	1.2702E+02	2.2951E-04	-9.6203E-05	6.0729E-06	-1.8605E-07	2.2807E-09	
R2	1.8838E-01	1.0818E-03	-1.3415E-04	-8.9339E-07	5.9204E-07	-3.1622E-08	
R3	1.5889E+02	3.3344E-04	-2.4609E-05	-8.4717E-07	1.3422E-07	-5.8371E-09	
R4	-2.2815E+00	1.2354E-04	-1.1096E-05	-6.8666E-07	6.5065E-08	-7.6475E-10	
R13	2.4141E+01	9.4874E-04	8.4097E-06	-2.2454E-07	8.9037E-09	-4.4577E-11	
R14	-1.3706E+02	8.2668E-04	4.3843E-05	4.9173E-07	-7.7032E-08	3.5685E-09	

Design data of the inflection point and the arrest point of each lens in the camera optical lens 40 according to Embodiment 4 of the present disclosure are shown in Tables 15 and 16.

TABLE 15

	Number of inflection points	Inflection point 1
P1R1	1	2.005
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	1	1.375

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TABLE 16

	Number of arrest points	Arrest point1
P1R1	1	3.265
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	1	2.305

FIG. 14 and FIG. 15 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 40 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 40 according to Embodiment 4, respectively. FIG. 16 is a schematic diagram of a field curvature and a distortion of the camera optical lens 40 after light having a wavelength of 555 nm passes through the camera optical lens 40 according to Embodiment 4. A field curvature S in FIG. 16 is a field curvature in a sagittal direction, and T is a field curvature in a meridian direction.

Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens 40 of this embodiment satisfies the above conditions.

In this embodiment, the camera optical lens 40 has an entrance pupil diameter ENPD of 3.911 mm, a full-field image height IH of 5.145 mm, and a field of view FOV 102.20° in a diagonal direction. The camera optical lens 40 has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

Embodiment 5

FIG. 17 is a structural schematic diagram of a camera optical lens 50 according to Embodiment 5 of the present disclosure. Embodiment 5 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

In this embodiment, the second lens L2 has a negative refractive power.

Table 17 and Table 18 show the design data of the camera optical lens 50 of the Embodiment 5 of the present disclosure.

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TABLE 17

	R	d	nd	vd
R1	58.524	d1=	1.000 nd1	1.8100 v1
R2	6.394	d2=	2.739	
R3	-26.960	d3=	1.611 nd2	1.8100 v2
R4	-10.562	d4=	-0.200	
S1	∞	d0=	0.704	
R5	-5.787	d5=	0.806 nd3	1.6204 v3
R6	14.553	d6=	0.000	
R7	14.553	d7=	4.500 nd4	1.9537 v4
R8	-11.272	d8=	0.134	
R9	14.627	d9=	3.288 nd5	1.6921 v5
R10	-11.087	d10=	0.000	
R11	-11.087	d11=	0.806 nd6	1.9229 v6
R12	26.794	d12=	0.600	
R13	14.641	d13=	4.000 nd7	1.5891 v7
R14	-10.000	d14=	0.403	
R15	∞	d15=	0.400 ndg	1.5168 vg
R16	∞	d16=	9.209	

Table 18 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 50 according to Embodiment 5 of the present disclosure.

TABLE 18

	Conic coefficient	Aspheric coefficient				
	k	A4	A6	A8	A10	A12
R1	1.2702E+02	1.5541E-03	-8.3040E-05	4.1280E-06	-1.0898E-07	8.3786E-10
R2	2.3390E+00	1.5610E-03	-9.4194E-05	9.4331E-06	-7.9449E-07	5.7644E-08
R3	5.2620E+01	-4.8282E-04	1.7696E-04	-2.1534E-05	2.4182E-06	-4.3223E-08
R4	7.3128E-01	-8.7169E-04	1.0347E-04	-2.2897E-05	2.7479E-06	-9.6706E-08
R13	-1.0927E+01	-4.7085E-05	-9.1800E-06	8.8604E-08	1.2889E-09	-1.2247E-10
R14	-2.7369E+00	-1.8288E-04	-3.6583E-06	4.0908E-08	-4.2052E-10	-4.6208E-11

Design data of the inflection point and the arrest point of each lens in the camera optical lens 50 according to Embodiment 5 of the present disclosure are shown in Tables 19 and 20.

TABLE 19

	Number of inflection points	Inflection point 1
P1R1	0	/
P1R2	0	/
P2R1	1	2.255
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	1	3.385
P7R2	0	/

TABLE 20

	Number of arrest points	Arrest point1
P1R1	0	/
P1R2	0	/
P2R1	1	2.945
P2R2	0	/
P3R1	0	/
P3R2	0	/

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TABLE 20-continued

	Number of arrest points	Arrest point1
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	1	5.195
P7R2	0	/

FIG. 18 and FIG. 19 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 50 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 50 according to Embodiment 5, respectively. FIG. 20 is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens 50 according to Embodiment 5. The field curvature S in FIG. 20 is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

Embodiment 6

Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens 50 of this embodiment satisfies the above conditions.

In this embodiment, the camera optical lens 50 has an entrance pupil diameter ENPD of 3.943 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 100.80° in a diagonal direction. The camera optical lens 50 has good optical performance such as large aperture, ultra-thinness and wide angle while taking into account characteristics of miniaturization, small chromatic aberration, small lateral color, high resolution up to 5M, and also has a long back focal length for easy assembling and low cost.

FIG. 21 is a structural schematic diagram of a camera optical lens 30 according to Embodiment 6 of the present disclosure. Embodiment 6 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

In this embodiment, the object-side surface of the second lens L2 is convex in the paraxial region, and the image-side surface of the second lens L2 is concave in the paraxial region.

Design data of the camera optical lens 60 of Embodiment 6 of the present disclosure are shown in Tables 21 and 22.

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TABLE 21

	R	d	nd	vd
R1	50.000	d1=	1.000	nd1 1.7433 v1 49.34
R2	4.156	d2=	2.000	
R3	17.804	d3=	1.611	nd2 1.7433 v2 49.40
R4	46.943	d4=	1.000	
S1	∞	d0=	1.800	
R5	-34.352	d5=	0.806	nd3 1.5174 v3 52.19
R6	21.623	d6=	0.000	
R7	21.623	d7=	2.551	nd4 1.9537 v4 32.32
R8	-9.900	d8=	0.133	
R9	64.025	d9=	3.974	nd5 1.8040 v5 46.57
R10	-18.191	d10=	0.000	
R11	-18.191	d11=	0.806	nd6 1.9229 v6 18.90
R12	10.317	d12=	0.300	
R13	9.576	d13=	3.370	nd7 1.6180 v7 63.41
R14	-8.807	d14=	0.403	
R15	∞	d15=	0.400	ndg 1.5168 vg 64.21
R16	∞	d16=	9.530	

Table 22 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 60 according to Embodiment 6 of the present disclosure.

TABLE 22

	Conic coefficient	Aspheric coefficient				
	k	A4	A6	A8	A10	A12
R1	9.4237E+01	3.8021E-04	-1.6748E-05	-1.4253E-06	7.5532E-08	-1.3466E-09
R2	1.5000E-01	8.2072E-04	6.2833E-05	-2.7321E-06	6.4190E-07	-6.8938E-08
R3	2.3873E+01	1.0533E-03	9.4246E-05	5.6581E-06	-5.1195E-07	2.0080E-08
R4	2.3909E+02	1.1012E-03	1.0540E-04	-1.8156E-06	3.9743E-08	3.5081E-08
R13	-1.6031E+00	-5.8426E-07	5.0444E-06	-2.0656E-07	-1.6793E-09	6.3518E-10
R14	-2.4235E+00	-1.3177E-04	-6.9161E-06	1.0693E-06	-6.8456E-08	1.9823E-09

Design data of the inflection point and the arrest point of each lens in the camera optical lens 60 according to Embodiment 6 of the present disclosure are shown in Tables 23 and 24.

TABLE 23

	Number of inflection points	Inflection point 1	Inflection point 2
P1R1	2	3.305	4.715
P1R2	0	/	/
P2R1	0	/	/
P2R2	0	/	/
P3R1	0	/	/
P3R2	0	/	/
P4R1	0	/	/
P4R2	0	/	/
P5R1	0	/	/
P5R2	0	/	/
P6R1	0	/	/
P6R2	0	/	/
P7R1	0	/	/
P7R2	0	/	/

TABLE 24

	Number of arrest points	Arrest point1
P1R1	0	/
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/

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TABLE 24-continued

	Number of arrest points	Arrest point1
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	0	/

FIG. 22 and FIG. 23 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 60 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 60 according to Embodiment 6, respectively. FIG. 24 is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens 60 according to Embodiment 6. The field curvature S in FIG. 24 is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

Comparative Example 1

FIG. 25 is a structural schematic diagram of a camera optical lens 70 according to Comparative Example 1 of the present disclosure. Comparative Example 1 is basically the same as Embodiment 1, and involves symbols having the same meanings as Embodiment 1, and only differences therebetween are listed below.

In Comparative Example 1, the second lens L2 has a negative refractive power. The object-side surface of the second lens L2 is convex in the paraxial region, and the image-side surface of the second lens L2 is concave in the paraxial region.

Design data of the camera optical lens 70 of Comparative Example 1 of the present disclosure are shown in Tables 25 and 26.

TABLE 25

	R		d	nd	vd		
R1	50.000	d1=	1.000	nd1	1.7725	v1	49.61
R2	5.323	d2=	2.000				
R3	16.938	d3=	1.611	nd2	1.7433	v2	49.40
R4	49.397	d4=	1.000				
S1	∞	d0=	1.000				
R5	-6.112	d5=	0.806	nd3	1.5174	v3	52.19
R6	18.922	d6=	0.000				
R7	18.922	d7=	2.551	nd4	1.9537	v4	32.32
R8	-9.689	d8=	0.133				
R9	15.061	d9=	3.974	nd5	1.6921	v5	54.57
R10	-18.419	d10=	0.000				
R11	-18.419	d11=	0.806	nd6	1.9229	v6	18.90
R12	11.079	d12=	0.300				
R13	8.745	d13=	3.370	nd7	1.6180	v7	63.41
R14	-7.833	d14=	0.403				
R15	∞	d15=	0.400	ndg	1.5168	vg	64.21
R16	∞	d16=	8.580				

Table 26 shows aspheric surface data of the first lens L1, the second lens L2 and the seventh lens L7 in the camera optical lens 70 of the Comparative Example 1 of the present disclosure.

TABLE 26

	Conic coefficient	Aspheric coefficient				
	k	A4	A6	A8	A10	A12
R1	8.8071E+01	6.7896E-04	-2.1167E-05	3.3667E-08	1.3424E-08	-5.1776E-10
R2	4.7917E-01	4.9019E-04	-2.9033E-06	2.9396E-06	-3.6912E-07	3.0765E-08
R3	-5.9390E+00	1.3867E-04	1.8942E-05	3.0625E-06	2.4364E-07	6.7089E-09
R4	2.4263E+02	2.3132E-05	9.8727E-06	1.7878E-06	1.7838E-07	1.9753E-08
R13	-2.7427E+00	-4.8109E-05	7.0907E-07	-1.8973E-08	2.5524E-10	-6.0620E-11
R14	-1.9943E+00	-5.5232E-05	-6.9793E-06	1.7321E-07	3.5475E-09	-2.4854E-10

Design data of the inflection point and the arrest point of each lens in the camera optical lens 70 according to Comparative Example 1 of the present disclosure are shown in Tables 27 and 28

TABLE 27

	Number of inflection points	Inflection point 1
P1R1	0	/
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	1	4.805
P7R2	0	/

TABLE 28

	Number of arrest points	Arrest point1
P1R1	0	/
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/

TABLE 28-continued

	Number of arrest points	Arrest point1
P4R2	0	/
P5R1	0	/
P5R2	0	/
P6R1	0	/
P6R2	0	/
P7R1	0	/
P7R2	0	/

FIG. 26 and FIG. 27 are schematic diagrams of a longitudinal aberration and a lateral color of the camera optical lens 70 after light having a wavelength of 650 nm, 610 nm, 555 nm, 510 nm and 470 nm passes through the camera optical lens 70 according to Comparative Example 1, respectively. FIG. 28 is a schematic diagram of a field curvature and a distortion after light having a wavelength of 555 nm passes through the camera optical lens 70 according to Comparative Example 1. The field curvature S in FIG. 28 is a field curvature in a sagittal direction, and T is a field curvature in a meridional direction.

Table 29 below lists the corresponding values of each condition in this embodiment according to the above conditions. It can be appreciated that the camera optical lens 70 of Comparative Example 1 does not satisfy the following condition: $1.30 \leq f/f_s \leq 4.00$.

In Comparative Example 1, the camera optical lens 70 has an entrance pupil diameter ENPD of 3.713 mm, a full-field image height IH of 5.145 mm, and a field of view FOV of 104.20° in a diagonal direction. The aberration of the camera optical lens 70 is not fully corrected, which is not beneficial to reducing the diameter of front opening, so that its optical performance is not excellent enough.

TABLE 29

Parameters and conditions	Embodiment 1	Embodiment 2	Embodiment 3	Embodiment 4	Embodiment 5	Embodiment 6	Comparative Example 1
f7/f	1.59	1.39	1.99	3.95	1.69	1.31	1.22
n1	1.77	1.77	1.77	1.88	1.81	1.74	1.77
f34/f	2.89	2.52	3.44	2.08	6.95	3.12	3.33
(FOV × f)/IH	125.79	124.81	124.92	124.31	124.09	121.09	120.31
f	6.320	6.289	6.264	6.258	6.334	6.096	5.940
f1	−9.750	−7.160	−6.358	−5.598	−8.899	−6.133	−7.760
f2	81.851	34.602	79.643	18.730	−33.138	37.558	−9.131
f3	−8.128	−13.169	7.409	−30.866	−6.554	−25.480	−8.801
f4	6.967	8.114	−10.255	9.909	7.241	7.373	6.986
f5	11.326	12.799	6.519	9.319	9.584	17.938	12.542
f6	−8.262	−8.200	−8.150	−8.998	−8.333	−6.972	−7.329
f7	10.020	8.741	12.442	24.715	10.701	7.960	7.229
FNO	1.600	1.600	1.600	1.600	1.606	1.700	1.600
TTL	30.004	28.180	27.530	29.015	30.000	29.684	27.934
IH	5.145	5.145	5.145	5.145	5.145	5.145	5.145
FOV	102.40°	102.10°	102.60°	102.20°	100.80°	102.20°	104.20°
D	4.617	4.763	4.545	4.384	4.884	5.009	4.956
f34	18.254	15.880	21.553	13.000	44.000	18.99	19.756

FOV denotes a field of view of the camera optical lens in a diagonal direction.

The above are only preferred embodiments of the present disclosure. Here, it should be noted that those skilled in the art may make modifications without departing from the inventive concept of the present disclosure, but these shall fall into the protection scope of the present disclosure.

What is claimed is:

1. A camera optical lens, comprising from an object side to an image side:

- a first lens;
- a second lens;
- a third lens;
- a fourth lens;
- a fifth lens;
- a sixth lens; and
- a seventh lens,

wherein the camera optical lens satisfies following conditions:

$$1.30 \leq f7/f \leq 4.00;$$

$$1.70 \leq n1;$$

$$2.00 \leq f34/f \leq 7.00; \text{ and}$$

$$120.00 \leq (FOV \times f)/IH$$

where

f denotes a focal length of the camera optical lens;
f7 denotes a focal length of the seventh lens;
f34 denotes a combined focal length of the third lens and the fourth lens;

n1 denotes a refractive index of the first lens;
FOV denotes a field of view of the camera optical lens; and

IH denotes an image height of the camera optical lens.

2. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition:

$$1.00 \leq f5/f \leq 3.00;$$

where f5 denotes a focal length of the fifth lens.

3. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions:

$$d10 = 0; \text{ and}$$

$$0.40 \leq |f5/f6| \leq 3.00,$$

where

d10 denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens;

f5 denotes a focal length of the fifth lens; and

f6 denotes a focal length of the sixth lens.

4. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition:

$$0.35 \leq BF/TL,$$

where

BF denotes an on-axis distance from an image-side surface of the seventh lens to an image plane of the camera optical lens; and

TL denotes an on-axis distance from an object-side surface of the first lens to the image-side surface of the seventh lens.

5. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition:

$$0.01 \leq (d4 + d0)/TTL \leq 0.10;$$

where

d4 denotes an on-axis distance from an image-side surface of the second lens to an aperture;

d0 denotes an on-axis distance from the aperture to an object-side surface of the third lens; and

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TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

6. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition:

$$0 < D/IH/FOV \leq 0.10;$$

10

where

D denotes a diameter of a light passing opening of an object-side surface of the first lens corresponding to a maximum field of view of the camera optical lens.

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7. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies a following condition:

$$0 < TTL/IH/FOV \leq 0.06;$$

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where TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

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8. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions:

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$$d6 = 0; \text{ and}$$

$$d10 = 0,$$

35

where

d6 denotes an on-axis distance from an image-side surface of the third lens to an object-side surface of the fourth lens; and

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d10 denotes an on-axis distance from an image-side surface of the fifth lens to an object-side surface of the sixth lens.

9. The camera optical lens as described in claim 1, wherein an object-side surface of the first lens is convex in a paraxial region, and an image-side surface of the first lens is concave in the paraxial region;

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the first lens has a negative refractive power; and

the camera optical lens satisfies following conditions:

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$$-3.09 \leq f1/f \leq -0.60;$$

$$0.57 \leq (R1 + R2)/(R1 - R2) \leq 1.88; \text{ and}$$

$$0.02 \leq d1/TTL \leq 0.05,$$

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where

f1 denotes a focal length of the first lens;

R1 denotes a central curvature radius of the object-side surface of the first lens;

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R2 denotes a central curvature radius of the image-side surface of the first lens;

d1 denotes an on-axis thickness of the first lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

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10. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions:

$$-10.46 \leq f2/f \leq 19.43;$$

$$-4.44 \leq (R3 + R4)/(R3 - R4) \leq 22.86; \text{ and}$$

$$0.03 \leq d3/TTL \leq 0.09,$$

where

f2 denotes a focal length of the second lens;

R3 denotes a central curvature radius of an object-side surface of the second lens;

R4 denotes a central curvature radius of an image-side surface of the second lens;

d3 denotes an on-axis thickness of the second lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

11. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions:

$$-9.86 \leq f3/f \leq 1.77;$$

$$-1.65 \leq (R5 + R6)/(R5 - R6) \leq 0.53; \text{ and}$$

$$0.01 \leq d5/TTL \leq 0.14,$$

where

f3 denotes a focal length of the third lens;

R5 denotes a central curvature radius of an object-side surface of the third lens;

R6 denotes a central curvature radius of an image-side surface of the third lens;

d5 denotes an on-axis thickness of the third lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

12. The camera optical lens as described in claim 1, wherein the camera optical lens satisfies following conditions:

$$-3.27 \leq f4/f \leq 2.38;$$

$$-0.14 \leq (R7 + R8)/(R7 - R8) \leq 1.27; \text{ and}$$

$$0.01 \leq d7/TTL \leq 0.23,$$

where

f4 denotes a focal length of the fourth lens;

R7 denotes a central curvature radius of an object-side surface of the fourth lens;

R8 denotes a central curvature radius of an image-side surface of the fourth lens;

d7 denotes an on-axis thickness of the fourth lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

13. The camera optical lens as described in claim 1, wherein an object-side surface of the fifth lens is convex in a paraxial region and an image-side surface of the fifth lens is convex in the paraxial region;

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the fifth lens has positive refractive power;
the camera optical lens satisfies following conditions:

$$\begin{aligned} -0.42 \leq (R9 + R10)/(R9 - R10) \leq 0.84; \text{ and} \\ 0.05 \leq d9/TTL \leq 0.22, \end{aligned}$$

where

R9 denotes a central curvature radius of an object-side surface of the fifth lens;

R10 denotes a central curvature radius of an image-side surface of the fifth lens;

d9 denotes an on-axis thickness of the fifth lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

14. The camera optical lens as described in claim **1**, wherein an object-side surface of the sixth lens is concave in a paraxial region;

the sixth lens has negative refractive power;
the camera optical lens satisfies following conditions:

$$\begin{aligned} -2.88 \leq f6/f \leq -0.76; \\ -2.78 \leq (R11 + R12)/(R11 - R12) \leq 0.42; \text{ and} \\ 0.01 \leq d11/TTL \leq 0.16, \end{aligned}$$

where

f6 denotes a focal length of the sixth lens;

R11 denotes a central curvature radius of an object-side surface of the sixth lens;

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R12 denotes a central curvature radius of an image-side surface of the sixth lens;

d11 denotes an on-axis thickness of the sixth lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

15. The camera optical lens as described in claim **1**, wherein an object-side surface of the seventh lens is convex in a paraxial region, and an image-side surface of the seventh lens is convex in the paraxial region;

the seventh lens has positive refractive power;

the camera optical lens satisfies following conditions:

$$\begin{aligned} -0.50 \leq (R13 + R14)/(R13 - R14) \leq 0.28; \text{ and} \\ 0.05 \leq d13/TTL \leq 0.20, \end{aligned}$$

where

R13 denotes a central curvature radius of an object-side surface of the seventh lens;

R14 denotes a central curvature radius of an image-side surface of the seventh lens;

d13 denotes an on-axis thickness of the seventh lens; and

TTL denotes a total optical length from an object-side surface of the first lens to an image plane of the camera optical lens along an optic axis.

16. The camera optical lens as described in claim **1**, wherein at least one of the first lens, the second lens, the third lens, the fourth lens, the fifth lens, the sixth lens, and the seventh lens is made of glass material.

* * * * *